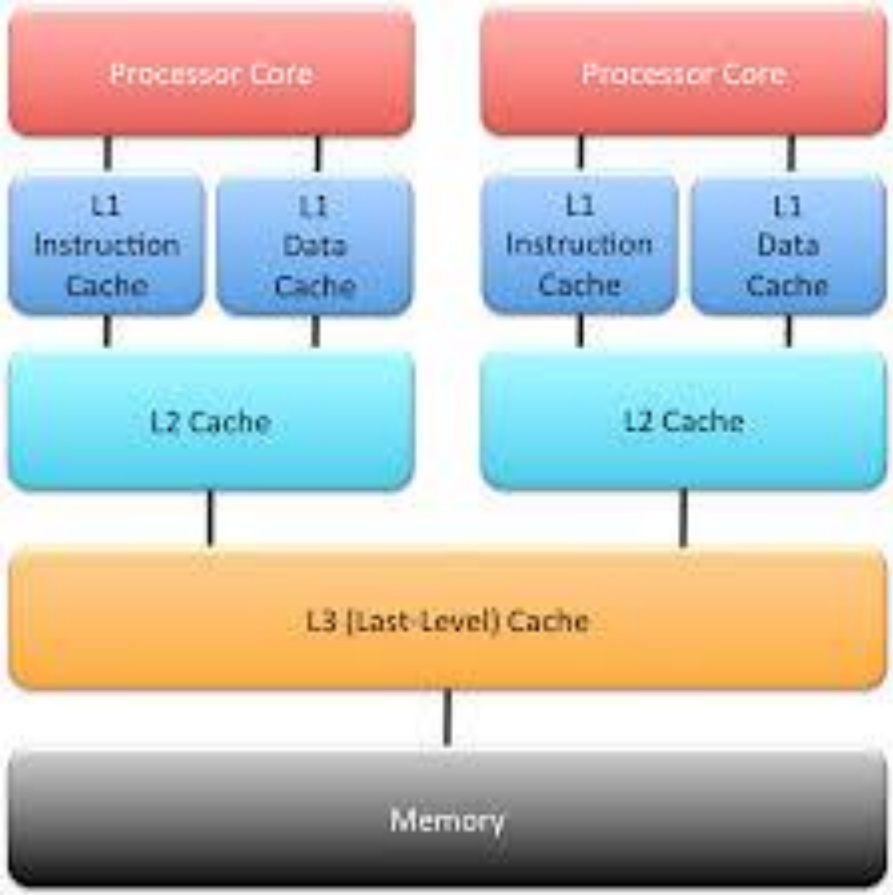
Unit3- Memory Management

- **Memory Management:** Memory management- address space abstraction- address binding-memory allocation- Fixed partition & variable partition memory strategies - dynamic address binding swapping- paging
- **virtual memory address translation**-dynamic paging static paging algorithms-dynamic paging algorithm-working set algorithms-segmentation-implementation-memory mapped files-concept of memory management in Linux & Windows NT/XP

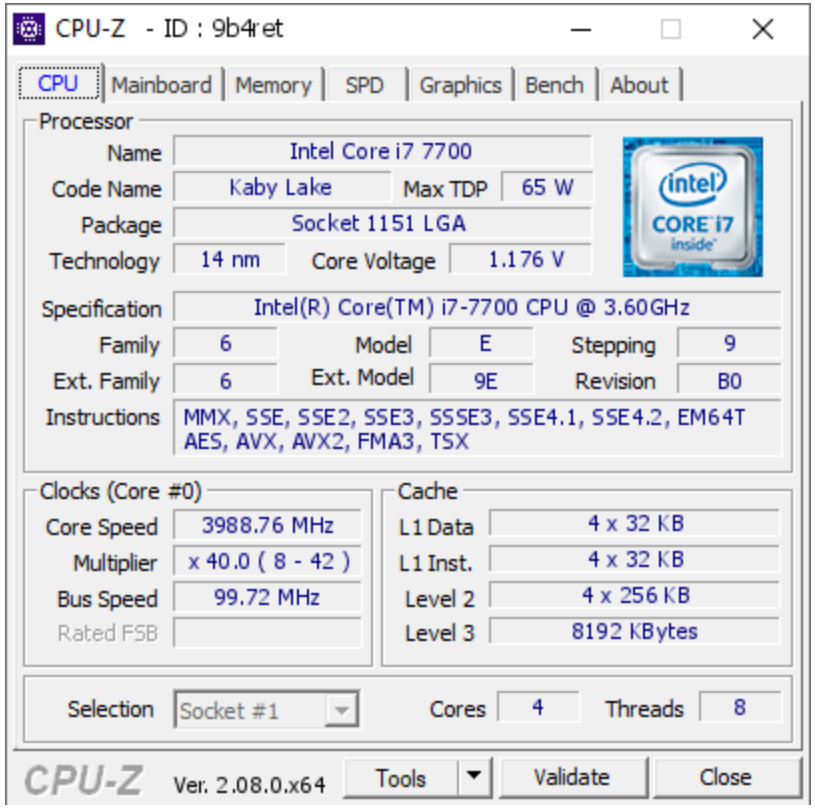
Background

- Program must be brought (from disk) into memory and placed within a process for it to be run
- Main memory and registers are only storage CPU can access directly
- Memory unit only sees a stream of addresses + read requests, or address + data and write requests
- Register access in one CPU clock (or less)
- Main memory can take many cycles, causing a stall
- Cache sits between main memory and CPU registers
- Protection of memory required to ensure correct operation

Cache Levels



CPU Specifications

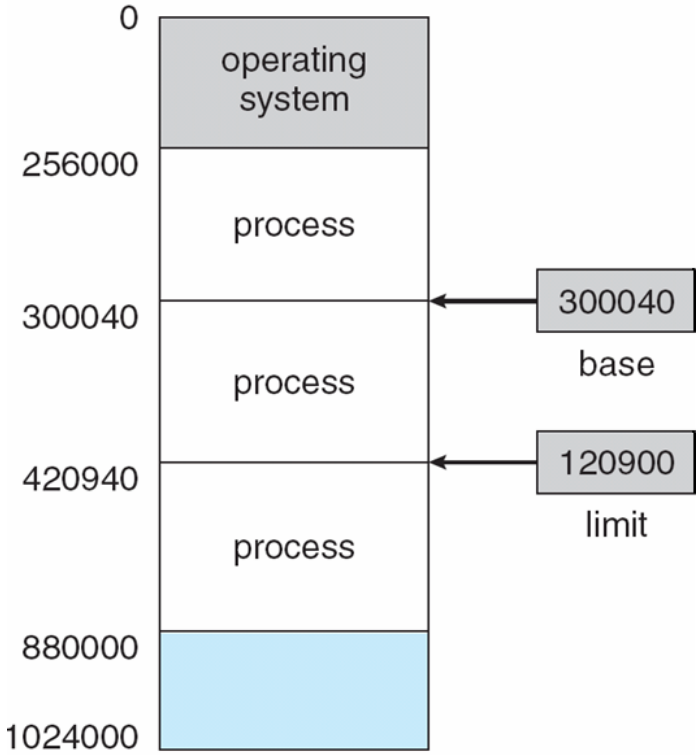


Background

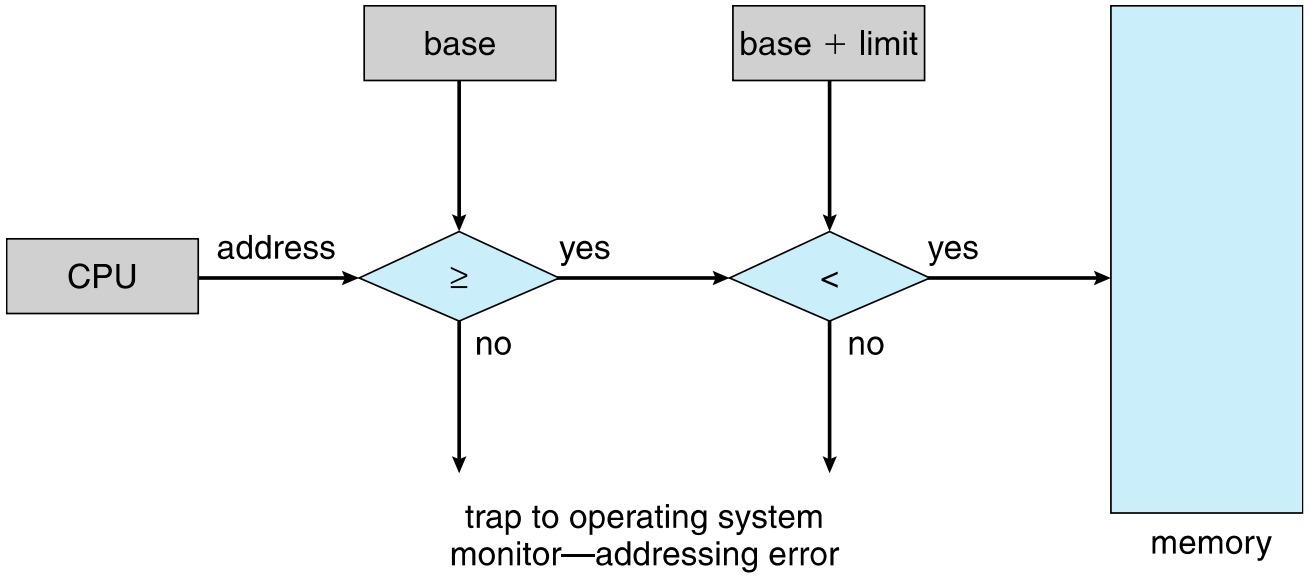
- ✓ We first need to make sure that each process has a separate memory space.
- ✓ Separate per-process memory space protects the processes from each other.
- ✓ We can provide this protection by using two registers, usually a base and a limit.

Base and Limit Registers

- A pair of **base** and **limit registers** define the logical address space
- CPU must check every memory access generated in user mode to be sure it is between base and limit for that user



Hardware Address Protection



The CPU hardware compares every address generated in user mode with the registers.
Any attempt by a program executing in user mode to access operating-system memory or other users' memory results in a trap to the operating system, which treats the attempt as a fatal error

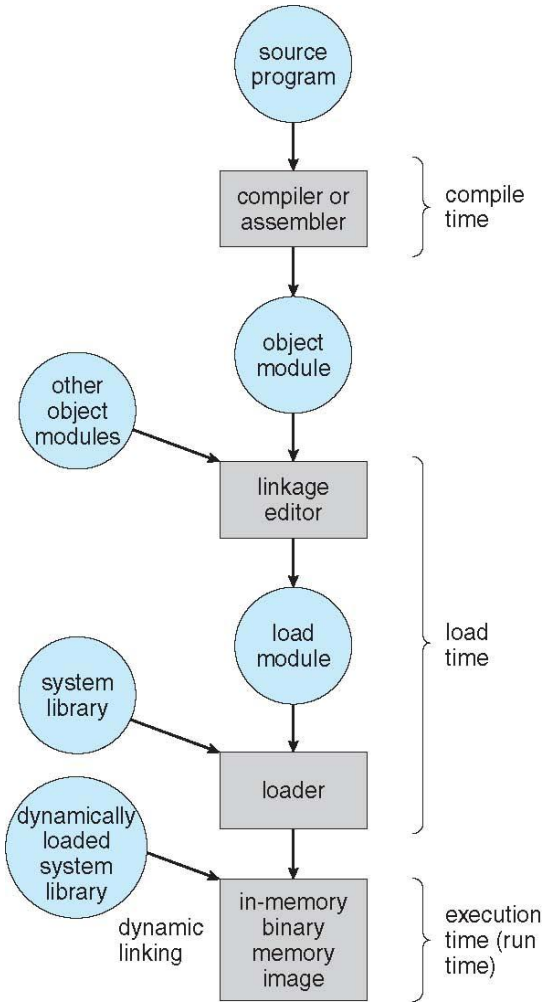
Address Binding

- Programs on disk, ready to be brought into memory to execute form an **input queue**
  - Without support, must be loaded into address 0000
- Inconvenient to have first user process physical address always at 0000
- Further, addresses represented in different ways at different stages of a program's life
  - Source code addresses are usually symbolic
  - A compiler typically binds these symbolic addresses to relocatable addresses (such as "14 bytes from the beginning of this module").
  - Linker or loader will bind relocatable addresses to absolute addresses
    - i.e. 74014
  - Each binding maps one address space to another

Binding of Instructions and Data to Memory

- Address binding of instructions and data to memory addresses can happen at three different stages:
  - **Compile time**: If memory location known a priori, **absolute code** can be generated; must recompile code if starting location changes
  - **Load time**: Must generate **relocatable code** if memory location is not known at compile time. Final binding is delayed until load time. If the starting address changes, we need only reload the user code to incorporate this changed value.
  - **Execution time**: Binding is delayed until run time if the process can be moved during its execution from one memory segment to another
    - Need hardware support for address maps (e.g., base and limit registers)
    - Most general-purpose operating systems use this method.

Multistep Processing of a User Program



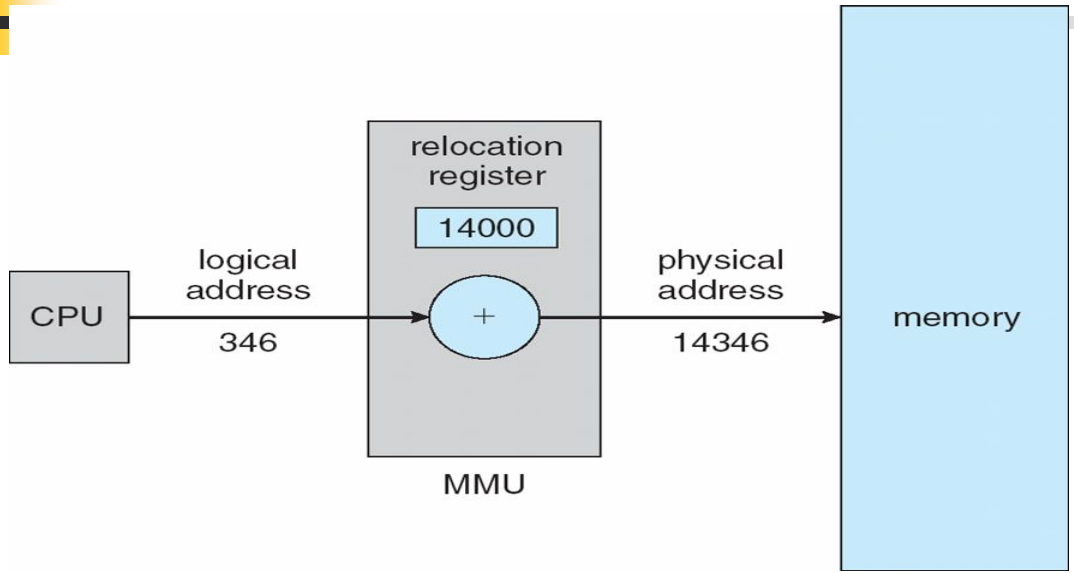
Logical vs. Physical Address Space

- The concept of a logical address space that is bound to a separate **physical address space** is central to proper memory management
  - **Logical address** – generated by the CPU; also referred to as **virtual address**
  - **Physical address** – address seen by the memory unit
- Logical and physical addresses are the same in compile-time and load-time. The execution-time address-binding scheme however results in differing logical and physical addresses.
- The run-time mapping from virtual to physical addresses is done by a hardware device called the memory-management unit (MMU).
- **Logical address space** is the set of all logical addresses generated by a program
- **Physical address space** is the set of all physical addresses generated by a program

Memory-Management Unit (MMU)

- Hardware device that maps virtual to physical address at run time.
- To start, consider simple scheme where the value in the relocation register is added to every address generated by a user process at the time it is sent to memory
  - Base register now called **relocation register**
  - (MS-DOS on Intel 80x86 used 4 relocation registers)
- The user program deals with *logical* addresses; it never sees the *real* physical addresses
  - Execution-time binding occurs when reference is made to location in memory
  - Logical address bound to physical addresses

Dynamic relocation using a relocation register



The value in the relocation register is added to every address generated by a user process at the time when the address is sent to memory. For example, if the base is at 14000, then an attempt by the user to address location 0 is dynamically relocated to location 14000; an access to location 346 is mapped to location 14346.

Logical vs. Physical Address

- The user program never sees the real physical addresses.
- logical addresses (in the range 0 to max)
- physical addresses (in the range R +0 to R + max for a base value R).
- The user program generates only logical addresses and thinks that the process runs in locations 0 to max.
- logical addresses must be mapped to physical addresses before they are used.

Dynamic Linking

- **Static linking** – system libraries and program code combined by the loader into the binary program image
- Dynamic linking –linking postponed until execution time
- Small piece of code, **stub**, used to locate the appropriate memory-resident library routine.
- Stub replaces itself with the address of the routine, and executes the routine
- Operating system checks if routine is present in processes' memory address
  - If not in address space, add to address space
- Dynamic linking is particularly useful for libraries

Memory Allocation

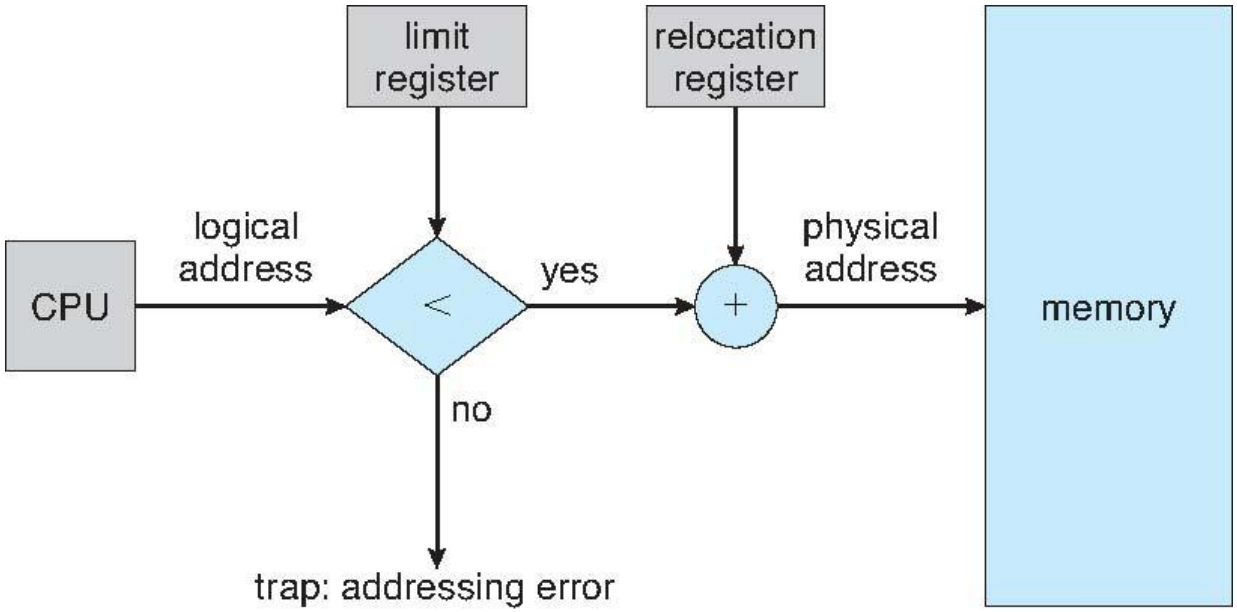
Contiguous Allocation

- Main memory must support both OS and user processes
- Limited resource, must allocate efficiently
- Contiguous allocation is one early method
- Main memory usually into two **partitions**:
  - Resident operating system, usually held in low memory with interrupt vector
  - User processes then held in high memory
  - Each process contained in single contiguous section of memory

Contiguous Allocation (Cont.)

- Relocation registers used to protect user processes from each other, and from changing operating-system code and data
  - Base register contains value of smallest physical address
  - Limit register contains range of logical addresses – each logical address must be less than the limit register
  - MMU maps logical address *dynamically*
  - Can then allow actions such as kernel code being **transient** and kernel changing size

Hardware Support for Relocation and Limit Registers

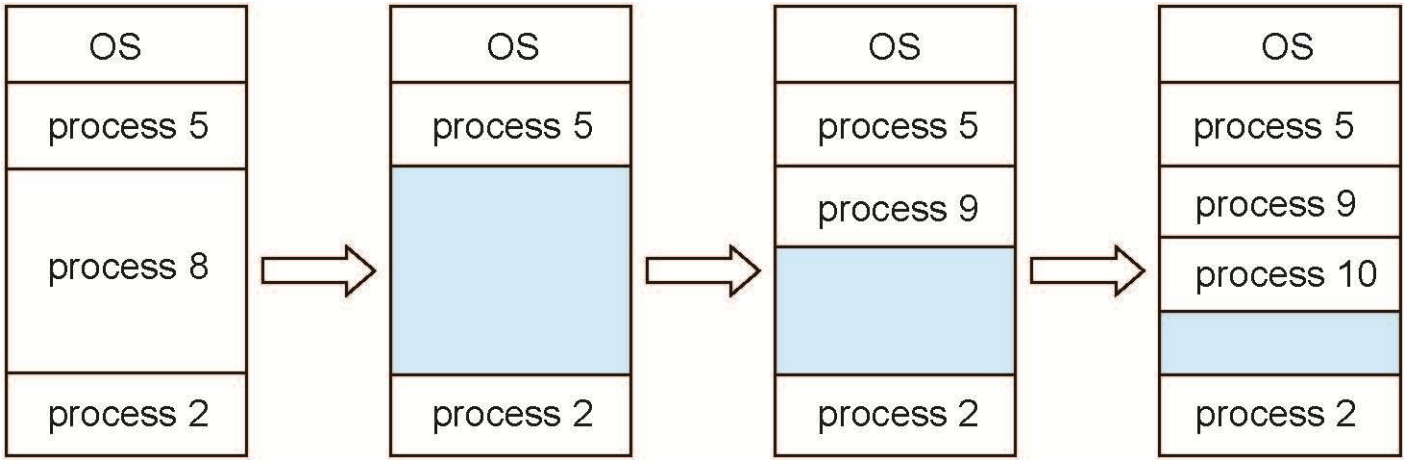


Fixed-size partition and Variable-size partition

- Multiple-partition allocation:
- **Fixed-size parttion:**
  - Each partition may contain exactly one process.
  - Degree of multiprogramming limited by number of partitions
  - In this multiple- partition method, when a partition is free, a process is selected from the input queue and is loaded into the free partition. When the process terminates, the partition becomes available for another process.
  - No longer used.
  - **Variable-partition:**
  - In the variable-partition scheme, the operating system keeps a table indicating which parts of memory are available and which are occupied.
  - Initially, all memory is available for user processes and is considered one large block of available memory, a hole.

Multiple-partition allocation

- Multiple-partition allocation
  - Degree of multiprogramming limited by number of partitions
  - **Variable-partition** sizes for efficiency (sized to a given process' needs)
  - **Hole** – block of available memory; holes of various size are scattered throughout memory
  - When a process arrives, it is allocated memory from a hole large enough to accommodate it
  - Process exiting frees its partition, adjacent free partitions combined
  - Operating system maintains information about:
    a) allocated partitions b) free partitions (hole)



Dynamic Storage-Allocation Problem

How to satisfy a request of size ***n*** from a list of free holes?

- **First-fit**: Allocate the ***first*** hole that is big enough
- **Best-fit**: Allocate the ***smallest*** hole that is big enough; must search entire list, unless ordered by size
  - Produces the smallest leftover hole
- **Worst-fit**: Allocate the ***largest*** hole; must also search entire list
  - Produces the largest leftover hole

First-fit and best-fit better than worst-fit in terms of speed and storage utilization

Practice Problem

Example 1:

Consider six memory partitions of size 200 KB, 400 KB, 600 KB, 500 KB, 300 KB and 250 KB. These partitions need to be allocated to four processes of sizes 357 KB, 210 KB, 468 KB and 491 KB in that order.

Perform the allocation of processes using:

1. First Fit Algorithm
2. Best Fit Algorithm
3. Worst Fit Algorithm

Example 2:

Given free memory partitions of 100K, 500K, 200K, 300K, and 600K (in order), how would each of the First-fit, Best-fit, and Worst-fit algorithms place processes of 212K, 417K, 112K, and 426K (in order)? Which algorithm makes the most efficient use of memory?

Fragmentation

- **External Fragmentation** – total memory space exists to satisfy a request, but it is not contiguous
- **Internal Fragmentation** – allocated memory may be slightly larger than requested memory; this size difference is memory internal to a partition, but not being used
- First fit analysis reveals that given *N* blocks allocated, 0.5 *N* blocks lost to fragmentation
  - 1/3 may be unusable -> **50-percent rule**

Fragmentation (Cont.)

- Reduce external fragmentation by **compaction**
  - Shuffle memory contents to place all free memory together in one large block.
  - If relocation is static and is done at assembly or load time, compaction cannot be done.
  - It is possible only if relocation is dynamic and is done at execution time.
  - Another possible solution to the external-fragmentation problem is to permit the logical address space of the processes to be noncontiguous

Segmentation

- Memory-management scheme that supports user view of memory.
- Segmentation permits the physical address space of a process to be non-contiguous.
- A program is a collection of segments
  - A segment is a logical unit such as:

    main program

    procedure

    function

    method

    object

    local variables, global variables

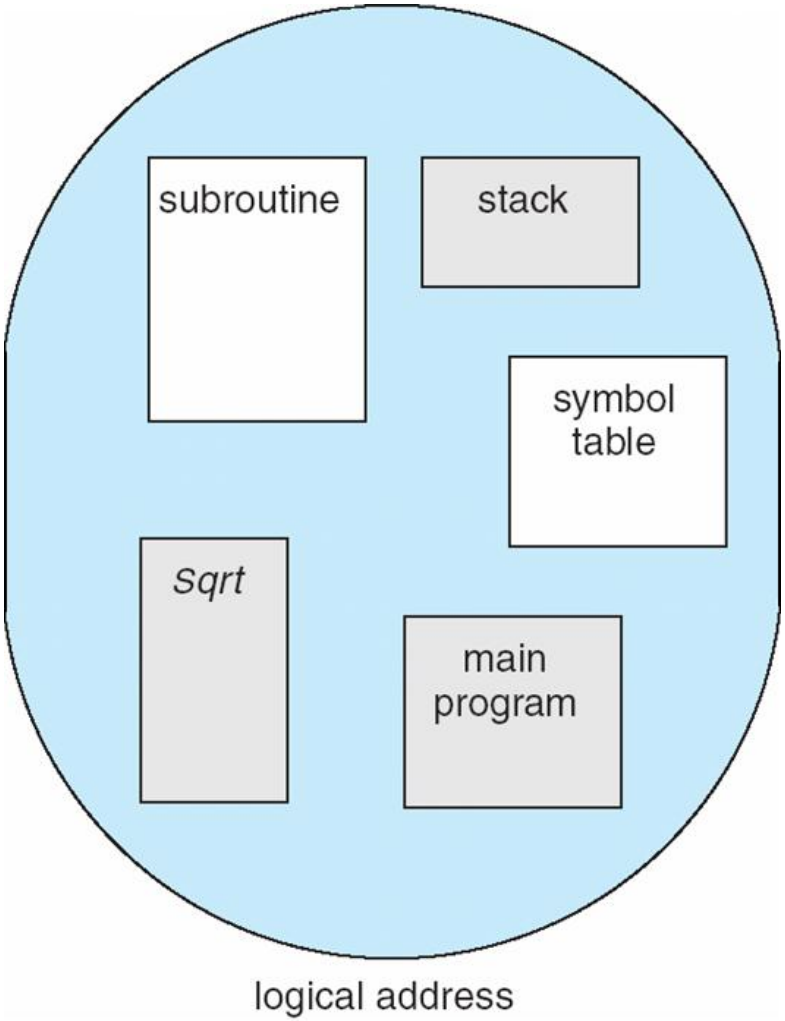
    common block

    stack

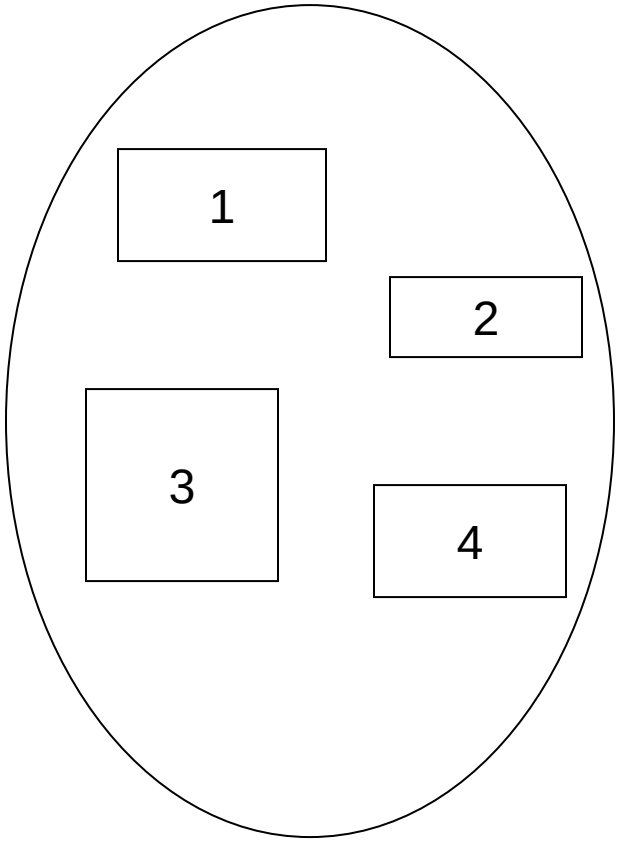
    symbol table

    arrays

User's View of a Program



Logical View of Segmentation



user space

physical memory space

Segmentation Architecture

- A logical address space is a collection of segments.
- Each segment has a name and a length. The addresses specify both the segment name and the offset within the segment.
- Logical address consists of a two tuple:

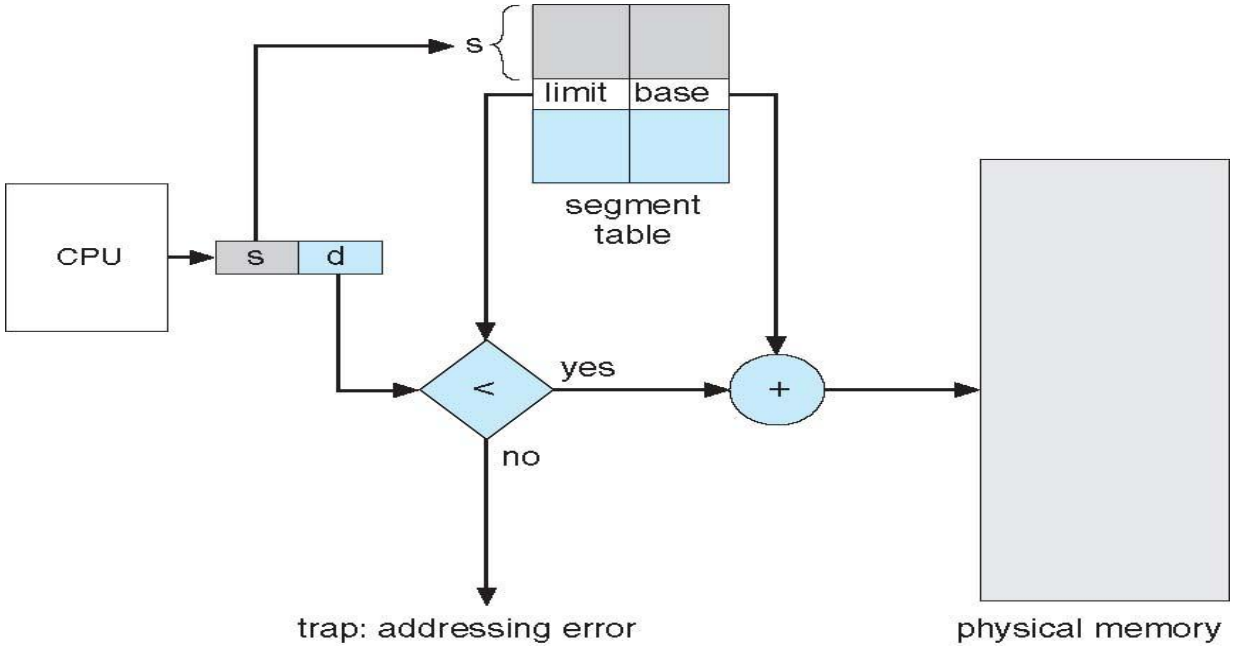
  <segment-number, offset>,

Mapping of two-dimensional user-defined addresses into one-dimensional physical is effected by a segment table.

- **Segment table** – maps two-dimensional physical addresses; each table entry has:
  - **base** – contains the starting physical address where the segments reside in memory
  - **limit** – specifies the length of the segment
- **Segment-table base register (STBR)** points to the segment table's location in memory
- **Segment-table length register (STLR)** indicates number of segments used by a program;

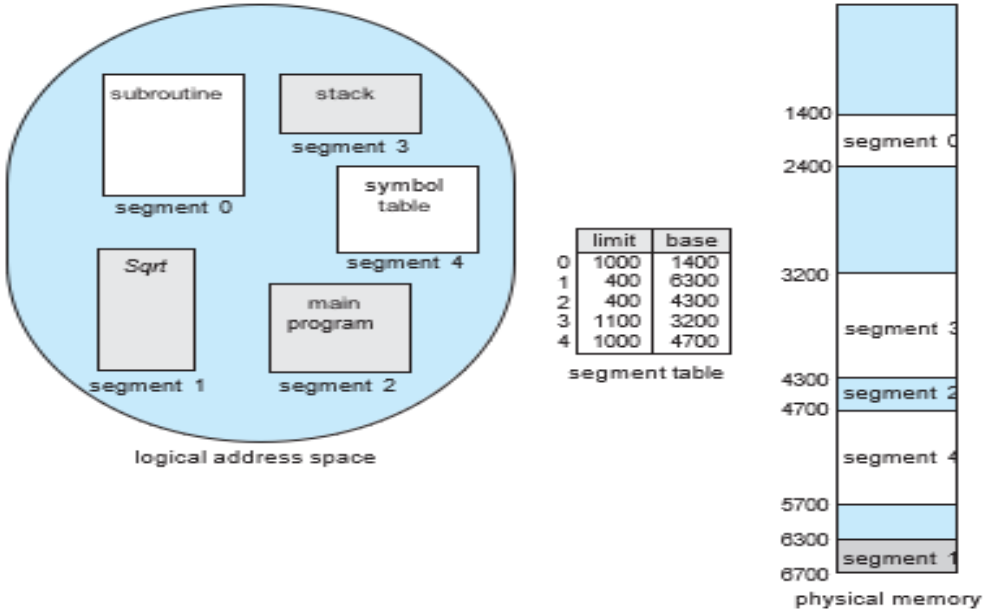
  segment number ***s*** is legal if ***s*** < **STLR**

Segmentation Hardware



A logical address consists of two parts: a segment number, *s*, and an offset into that segment, *d*. The segment number is used as an index to the segment table. The offset *d* of the logical address must be between 0 and the segment limit. If it is not, we trap to the operating system.
When an offset is legal, it is added to the segment base to produce the address in physical memory of the desired byte.

Example of segmentation



We have five segments numbered from 0 through 4. The segments are stored in physical memory as shown. The segment table has a separate entry for each segment, giving the beginning address of the segment in physical memory (or base) and the length of that segment (or limit). For example, segment 2 is 400 bytes long and begins at location 4300. Thus, a reference to byte 53 of segment 2 is mapped onto location 4300 + 53 = 4353. A reference to segment 3, byte 852, is mapped to 3200 (the base of segment 3) + 852 = 4052. A reference to byte 1222 of segment 0 would result in a trap to the operating system,.

Advantages and Disadvantages of Segmentation

Advantages

1. No internal fragmentation
2. Average Segment Size is larger than the actual page size.
3. Less overhead
4. It is easier to relocate segments than entire address space.
5. The segment table is of lesser size as compare to the page table in paging.

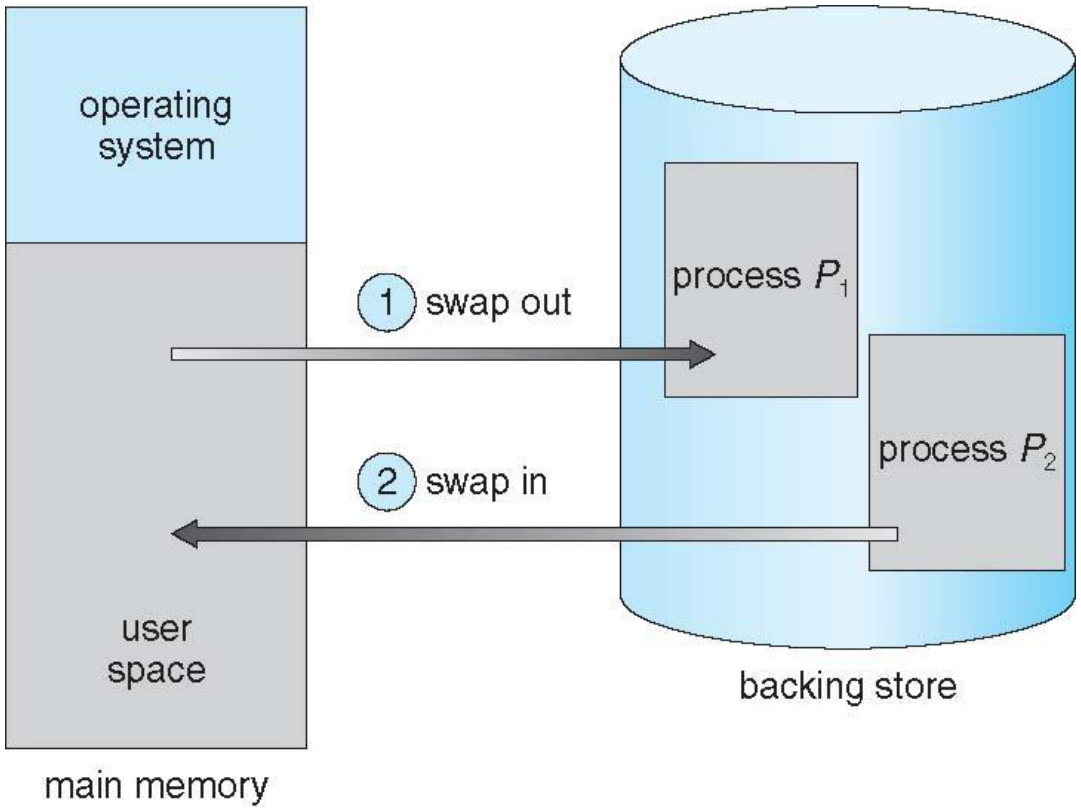
Disadvantages :

1. It can have external fragmentation.
2. It is difficult to allocate contiguous memory to variable sized partition.
3. Costly memory management algorithms

Swapping

- A process can be **swapped** temporarily out of memory to a backing store, and then brought back into memory for continued execution
  - Total physical memory space of processes can exceed physical memory
- **Backing store** – fast disk large enough to accommodate copies of all memory images for all users; must provide direct access to these memory images
- The system maintains a ready queue consisting of all processes whose memory images are on the backing store or in memory and are ready to run.
- Whenever the CPU scheduler decides to execute a process, it calls the dispatcher.
- Major part of swap time is transfer time; total transfer time is directly proportional to the amount of memory swapped

Schematic View of Swapping



Context Switch Time including Swapping

- If next processes to be put on CPU is not in memory, need to swap out a process and swap in target process
- Context switch time can then be very high
- 100MB process swapping to hard disk with transfer rate of 50MB/sec
  - Swap out time of 2000 ms
  - Plus swap in of same sized process
  - Total context switch swapping component time of 4000ms (4 seconds)
- Can reduce if reduce size of memory swapped – by knowing how much memory really being used
  - System calls to inform OS of memory use via `request_memory()` and `release_memory()`

Context Switch Time and Swapping (Cont.)

- Other constraints as well on swapping
  - A process may be waiting for an I/O operation when we want to swap that process to free up memory. However, if the I/O is asynchronously accessing the user memory for I/O buffers, then the process cannot be swapped.
  - There are two main solutions to this problem: never swap a process with pending I/O, or execute I/O operations only into operating-system buffers.
    - Known as **double buffering**, adds overhead
- Standard swapping not used in modern operating systems
  - But modified version common
    - Swap only when free memory extremely low

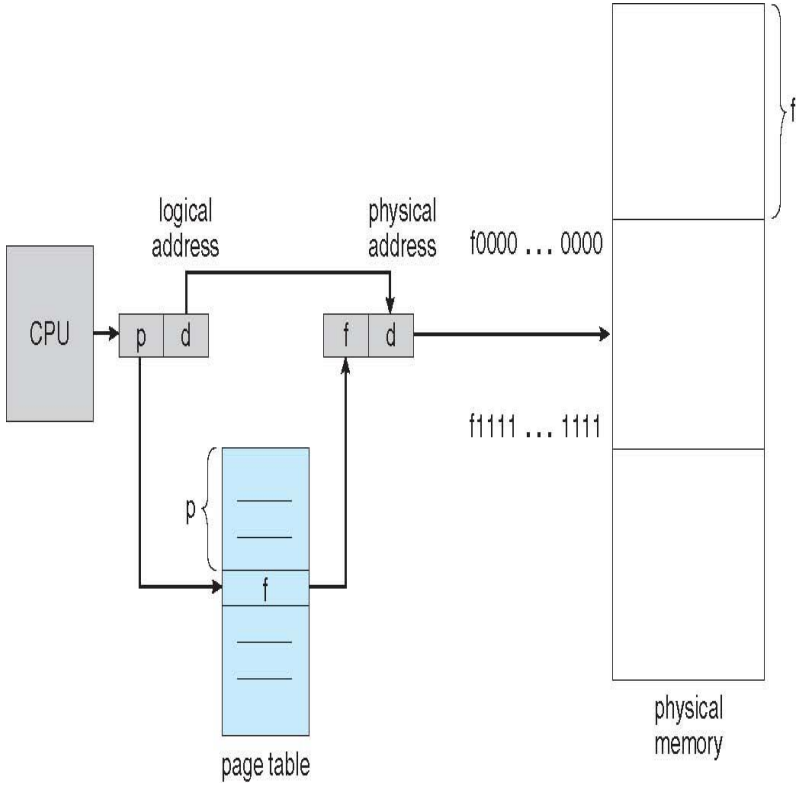
Paging

- **Paging** is memory-management scheme which ***permits the physical address space of a process to be non- contiguous.***
- ***Paging is implemented through cooperation between the operating system and the computer hardware.***
- Physical address space of a process can be noncontiguous; process is allocated physical memory whenever the latter is available
  - **Avoids external fragmentation and the need for compaction.**
  - Avoids problem of varying sized memory chunks
- Divide physical memory into fixed-sized blocks called **frames**
  - Size is power of 2, between 512 bytes and 16 Mbytes
- Divide logical memory into blocks of same size called **pages**
- Keep track of all free frames
- To run a program of size ***N*** pages, need to find ***N*** free frames and load program
- Set up a **page table** to translate logical to physical addresses
- Backing store likewise split into pages
- Still have Internal fragmentation

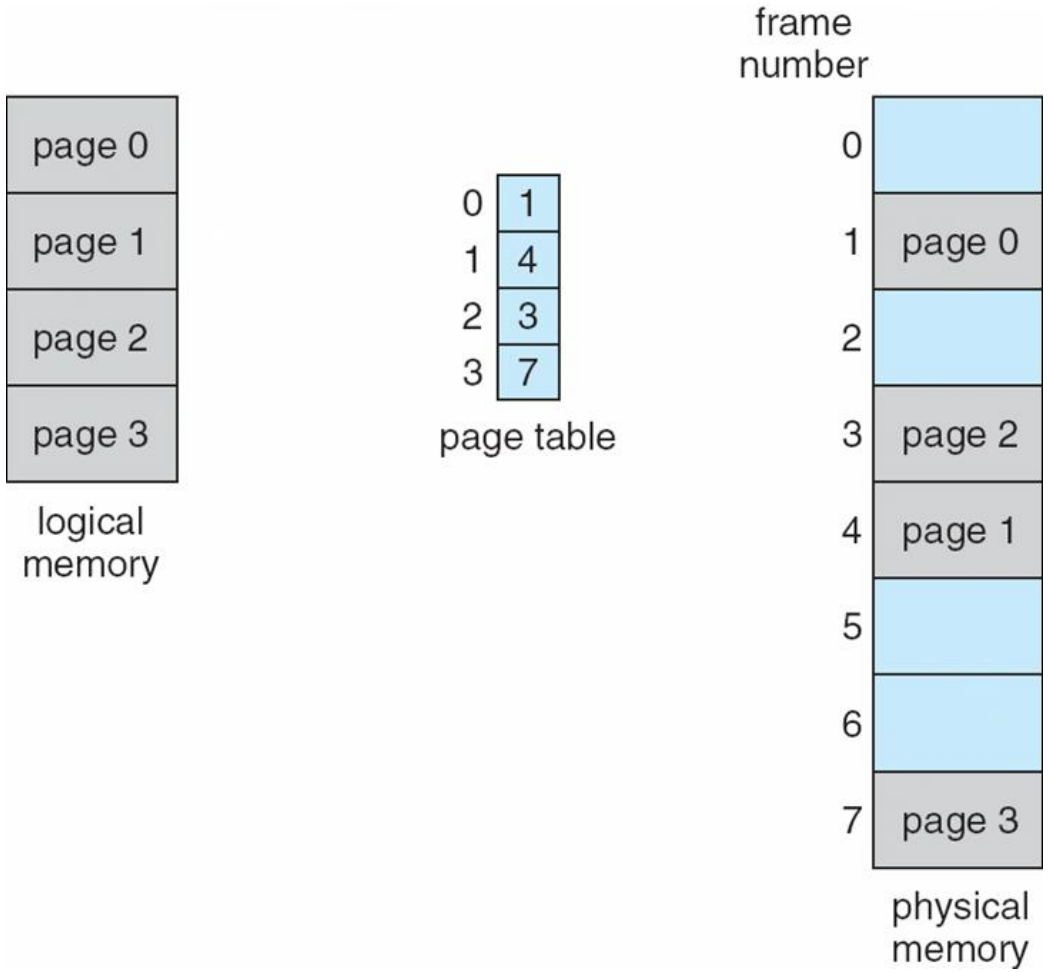
Paging Hardware

Address generated by CPU is divided into:

- **Page number** (***p***) – used as an index into a **page table** which contains base address of each page in physical memory
- **Page offset** (***d***) – combined with base address to define the physical memory address that is sent to the memory unit.



Paging Model of Logical and Physical Memory

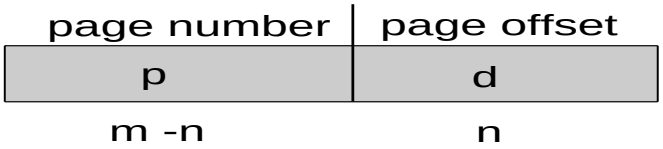


Address Translation Scheme

- The page size is power of 2, varying between 512 bytes and 1 GB per page is defined by the hardware.

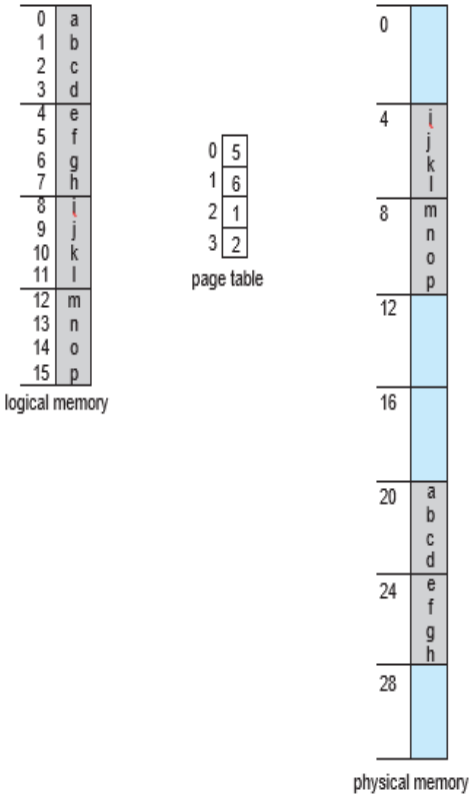
- If the size of the logical address space is 2^m, and a page size is 2^n bytes, then the high-order **m-n** bits of a logical address designate the page number, and the n low-order bits designate the page offset.

- The logical address is as follows:



where *p* is an index into the page table and *d* is the displacement within the page.

Paging example for a 32-byte memory with 4-byte pages



Consider the figure. Here in logical address, n= 2 and m = 4. Using a page size of 4 bytes and a physical memory of 32 bytes (8 pages), we show how the programmer's view of memory can be mapped into physical memory.

Logical address 0 is page 0, offset 0. Indexing into the page table, we find that page 0 is in frame 5. Thus, logical address 0 maps to physical address 20 [= (5 x 4) + 0].

Logical address 3 (page 0, offset 3) maps to physical address 23 [= (5 x 4) + 3]. Logical address 4 is page 1, offset 0; according to the page table, page 1 is mapped to frame 6

Paging (Cont.)

- Calculating internal fragmentation
  - Page size = 2,048 bytes
  - Process size = 72,766 bytes
  - 35 pages + 1,086 bytes
  - Internal fragmentation of 2,048 - 1,086 = 962 bytes
  - Worst case fragmentation = 1 frame – 1 byte
  - On average fragmentation = 1 / 2 frame size
  - So small frame sizes desirable?
  - Disk I/O is more efficient when the amount data being transferred is larger.
  - Page sizes growing over time
    - Solaris supports two page sizes – 8 KB and 4 MB

Paging (Cont.)

When we use a paging scheme, we **have no external fragmentation. Why?**

Any free frame can be allocated to a process that needs it.
However, we may have some internal fragmentation. When a process arrives in the system to be executed, its size, expressed in pages, is examined. Each page of the process needs one frame.

The first page of the process is loaded into one of the allocated frames, and the frame number is put in the page table for this process. The next page is loaded into another frame, its frame number is put into the page table, and so on.

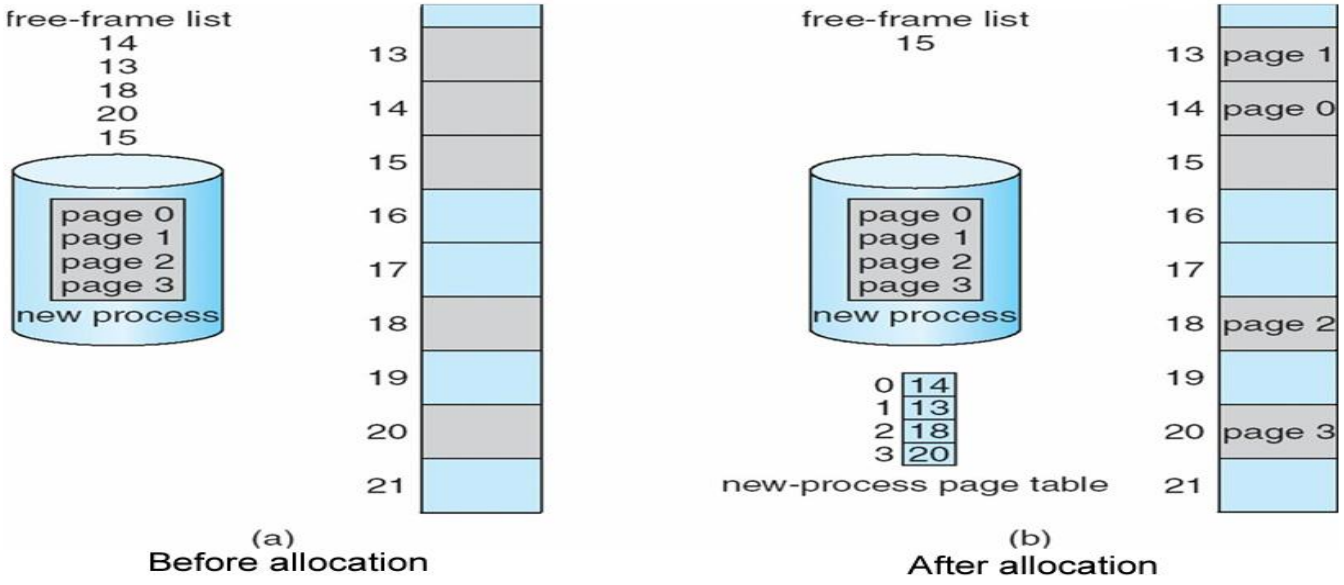
Overhead is involved in each page-table entry, and this overhead is reduced as the size of the pages increases. Also, disk I/O is more efficient when the amount data being transferred is larger.

Today, pages typically are between 4 KB and 8 KB in size, and some systems support even larger page sizes. Some CPUs and kernels even support multiple page sizes.

The operating system maintains a copy of the page table for each process.

Paging (Cont.)

When a process arrives in the system to be executed, its size, expressed in pages, is examined. Each page of the process needs one frame. Thus, if the process requires *n* pages, at least *n* frames must be available in memory. If *n* frames are available, they are allocated to this arriving process. The first page of the process is loaded into one of the allocated frames, and the frame number is put in the page table for this process. The next page is loaded into another frame, its frame number is put into the page table, and so on,



Frame Table

Since the operating system is managing physical memory, it must be aware of the allocation details of physical memory—which frames are allocated, which frames are available, how many total frames there are, and so on.

This information is generally kept in a data structure called a **frame table**.

The frame table has one entry for each physical page frame, indicating whether the latter is free or allocated and, if it is allocated, to which page of which process or processes.

Implementation of Page Table

- Page table is kept in main memory
- **Page-table base register** (**PTBR**) points to the page table
- **Page-table length register** (**PTLR**) indicates size of the page table
- ***In this scheme every data/instruction access requires two memory accesses:-Disadvantage***
  - One for the page table and one for the data / instruction. This will reduce the speed.
  - Even though many instruction belong to same page, each time we need to know the page number.
- The two memory access problem can be solved by the use of a special fast-lookup hardware cache called **associative memory** or **translation look-aside buffers** (**TLBs**)
- The TLB is associative, high-speed memory.

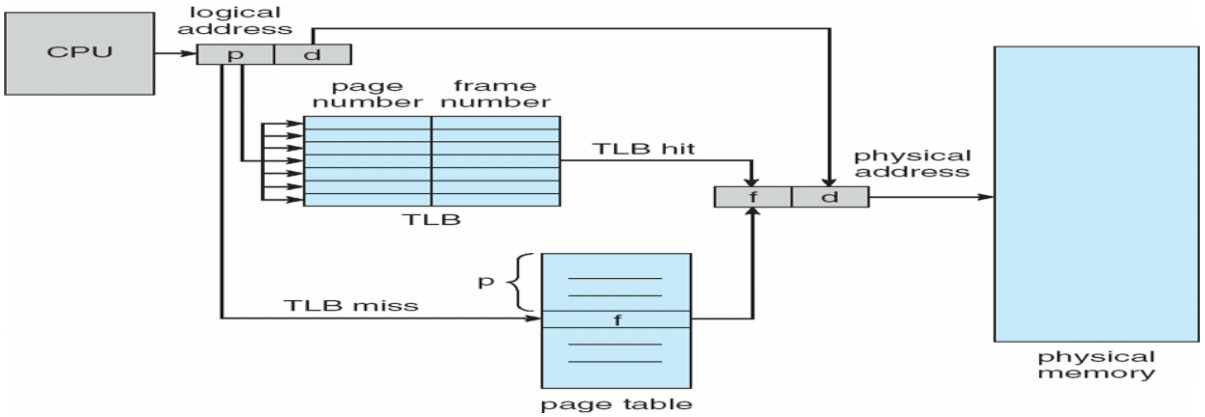
Implementation of Page Table (Cont.)

- Some TLBs store **address-space identifiers** (**ASIDs**) in each TLB entry – uniquely identifies each process to provide address-space protection for that process
  - Otherwise need to flush at every context switch
- TLBs typically small (64 to 1,024 entries)
- When a logical address is generated by the CPU, its page number is presented to the TLB. If the page number is found, its frame number is immediately available and is used to access memory.
- If the page number is not in the TLB (known as a **TLB miss**), a memory reference to the **page table** must be made.
- On a TLB miss, value is loaded into the TLB for faster access next time.
- If the TLB is already full of entries, an existing entry must be selected for replacement.

Associative Memory

- Replacement policies (LRU, round-robin) must be considered
- Some entries can be wired down(cannot be removed from the TLB) for permanent fast access.
- Some TLBs store address-space identifiers (ASIDs) in each TLB entry. An ASID uniquely identifies each process and is used to provide address-space protection for that process.
- **Hit ratio:** The percentage of times that the page number of interest is found in the TLB is called the hit ratio.

Paging Hardware With TLB



Effective Access Time calculations

- If it takes 100 nanoseconds to access memory, then a mapped-memory access takes 100 nanoseconds when the page number is in the TLB.
- An 80-percent hit ratio, for example, means that we find the desired page number in the TLB 80 percent of the time.
- If we **fail** to find the page number in the TLB then we must first access memory for the **page table** and frame number (100 nanoseconds) and then access the desired byte in memory (100 nanoseconds), for a total of 200 nanoseconds.

To find the effective memory-access time, we weight the case by its probability:

effective access time = 0.80x100 + 0.20x200

= 120 nanoseconds

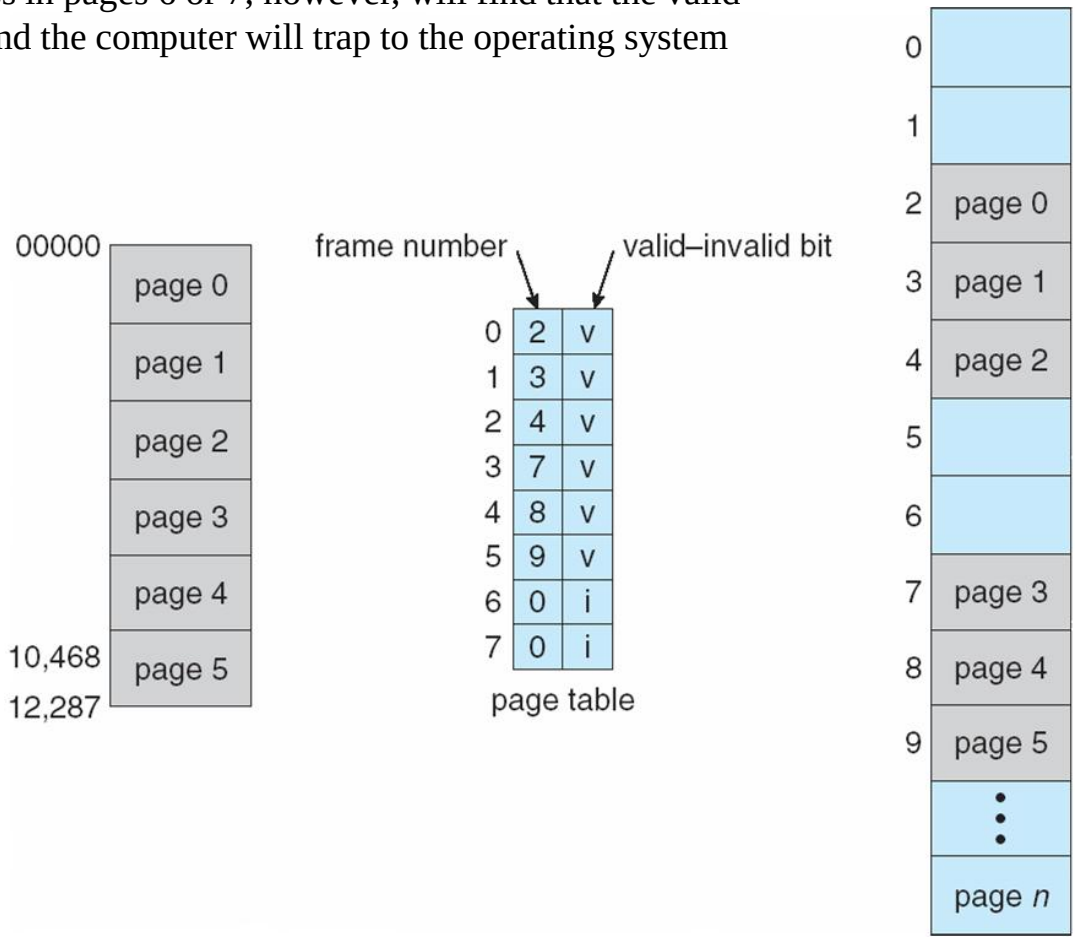
Example: For a 99-percent hit ratio calculate effective access time.

Memory Protection

- Memory protection implemented by associating protection bit with **each frame** to indicate if read-only or read-write access is allowed to that frame.
  - Can also add more bits to indicate page execute-only, and so on
- **Valid-invalid** bit attached to each entry in the page table:
  - "valid" indicates that the associated page is in the process's logical address space, and is thus a legal page
  - "invalid" indicates that the page is not in the process's logical address space
  - Wastage of valuable memory- many processes use only a small fraction of the address space available to them.
  - Use hardware, in the form of a **page-table length register (PTLR)**, to indicate the size of the page table.
  - Any violations result in a trap to the kernel

Valid (v) or Invalid (i) Bit In A Page Table

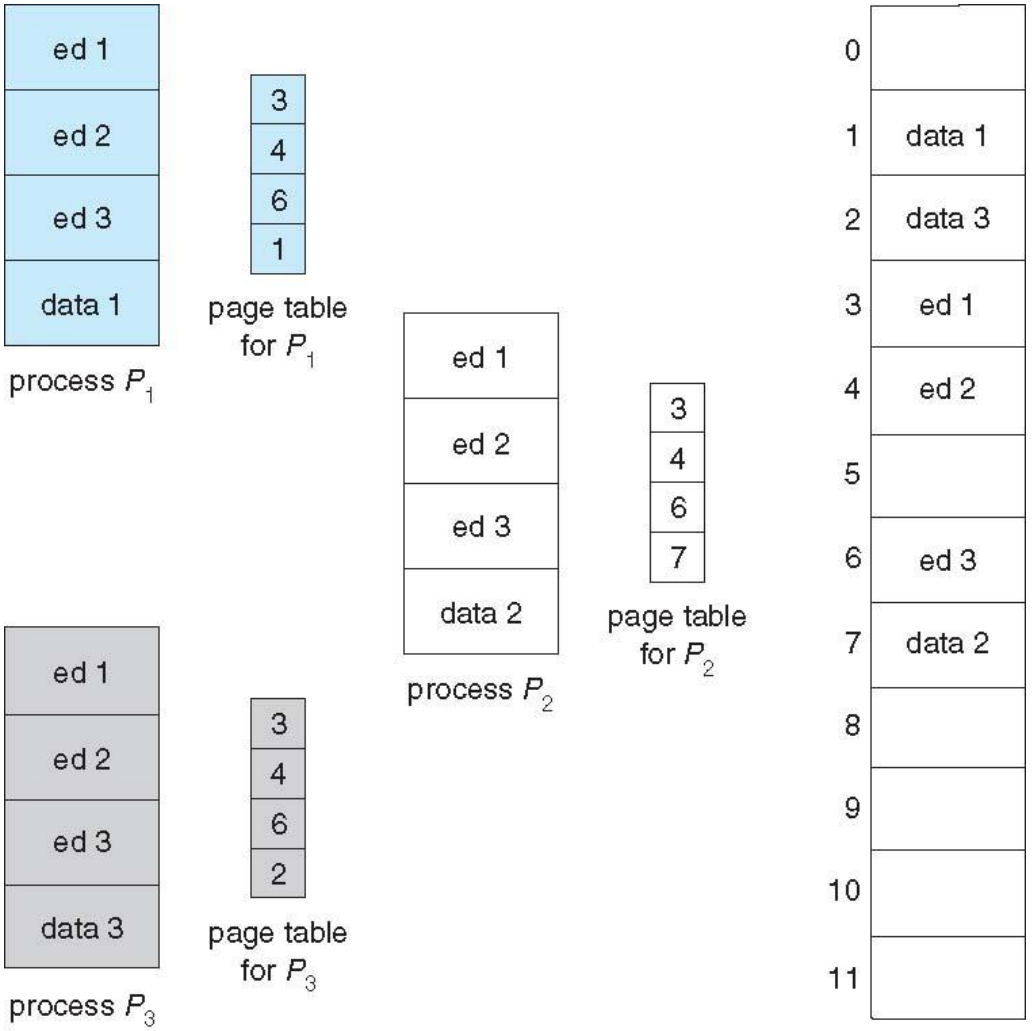
14-bit address space (0 to 16383),program uses only addresses 0 to 10468,
page size of 2 KB
attempt to generate an address in pages 6 or 7, however, will find that the valid–invalid bit is set to **invalid**, and the computer will trap to the operating system



Shared Pages

- **Shared code**
  - One copy of read-only (**reentrant - non-self modifying**) code shared among processes (i.e., text editors, compilers, window systems)
  - Similar to multiple threads sharing the same process space
  - Also useful for interprocess communication if sharing of read-write pages is allowed
- **Private code and data**
  - Each process keeps a separate copy of the code and data
  - The pages for the private code and data can appear anywhere in the logical address space

Shared Pages Example



Structure of the Page Table

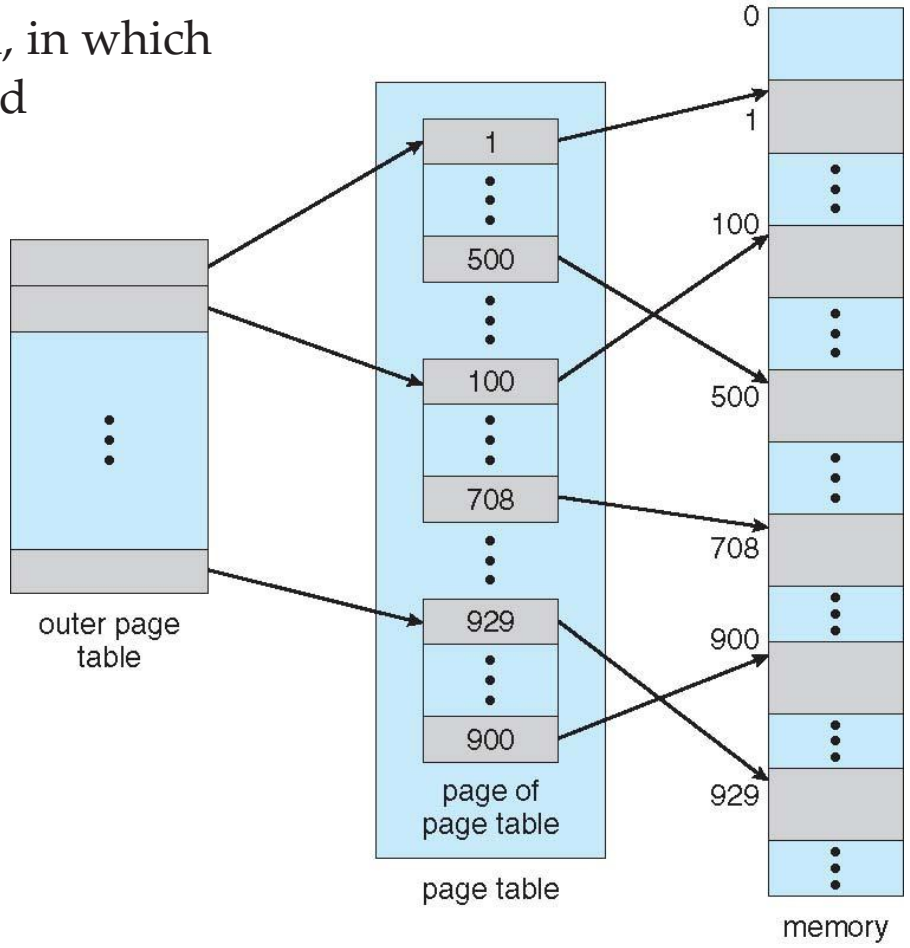
- Memory structures for paging can get huge using straight-forward methods
  - Consider a 32-bit logical address space as on modern computers
  - Page size of 4 KB (2^{12})
  - Page table would have 1 million entries ($2^{32} / 2^{12}$)
  - If each entry is 4 bytes -> 4 MB of physical address space / memory for page table alone
    - That amount of memory used to cost a lot
    - **Don't want to allocate that contiguously in main memory**
    - **Divide the page table into smaller pieces.**
- Hierarchical Paging
- Hashed Page Tables
- Inverted Page Tables

Hierarchical Page Tables

- Break up the logical address space into multiple page tables
- A simple technique is a two-level page table
- We then page the page table

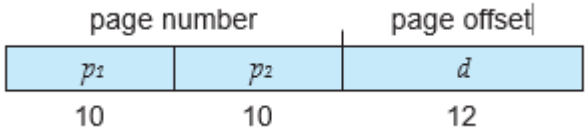
Two-Level Page-Table Scheme

Use two-level paging algorithm, in which the page table itself is also paged

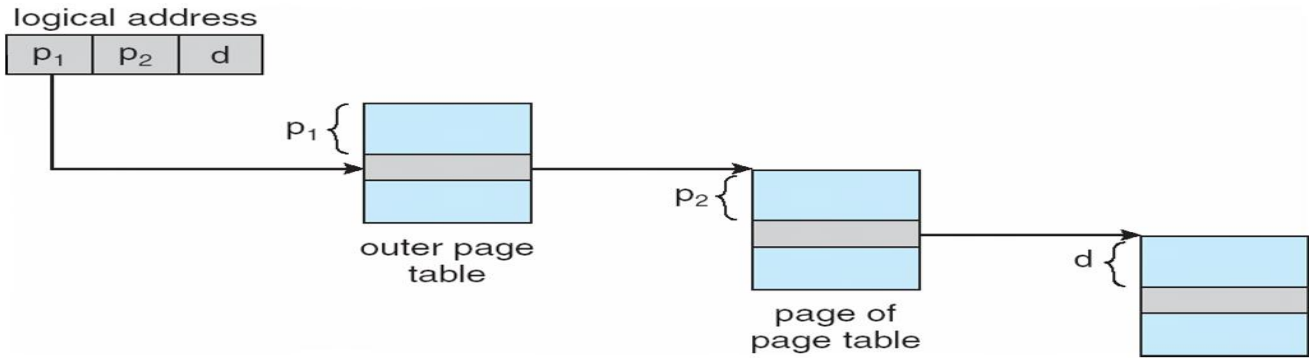


Address translation for a two-level 32-bit paging architecture

For example, consider again the system with a 32-bit logical address space and a page size of 4 KB. A logical address is divided into a page number consisting of 20 bits and a page offset consisting of 12 bits. Because we page the page table, the page number is further divided into a 10-bit page number and a 10-bit page offset. Thus, a logical address is as follows:



where p_1 is an index into the outer page table and p_2 is the displacement within the page of the inner page table. The address-translation method for this architecture is shown in in following figure:



Virtual Memory-Background

- Virtual memory is a technique that allows the execution of processes that are not completely in memory.
- Code needs to be in main memory to execute, but entire program rarely used
  - Error code, unusual routines, large data structures (Arrays, matrices)
- Entire program code **not** needed at same time
- Consider ability to execute partially-loaded program
  - Program no longer constrained by limits of physical memory
  - Each program takes less memory while running -> more programs run at the same time
    - Increased CPU utilization and throughput with no increase in response time or turnaround time
  - Less I/O needed to load or swap programs into memory -> each user program runs faster

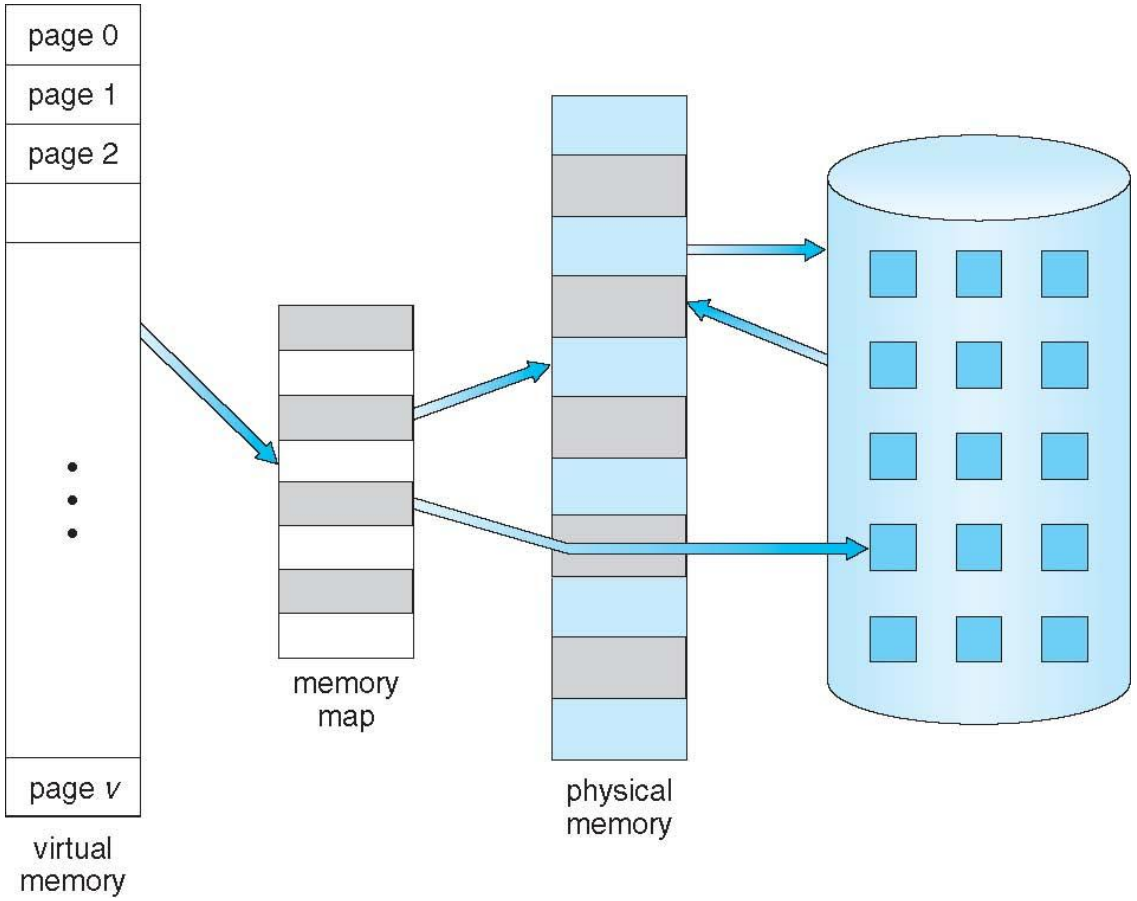
Virtual Memory-Background

- **Virtual memory** – separation of user logical memory from physical memory
  - Only part of the program needs to be in memory for execution
  - Logical address space can therefore be much larger than physical address space
  - Allows address spaces to be shared by several processes
  - Allows for more efficient process creation
  - More programs running concurrently
  - Less I/O needed to load or swap processes
  - Virtual memory makes the task of programming much easier, because the programmer no longer needs to worry about the amount of physical memory available.

Virtual Memory-Background

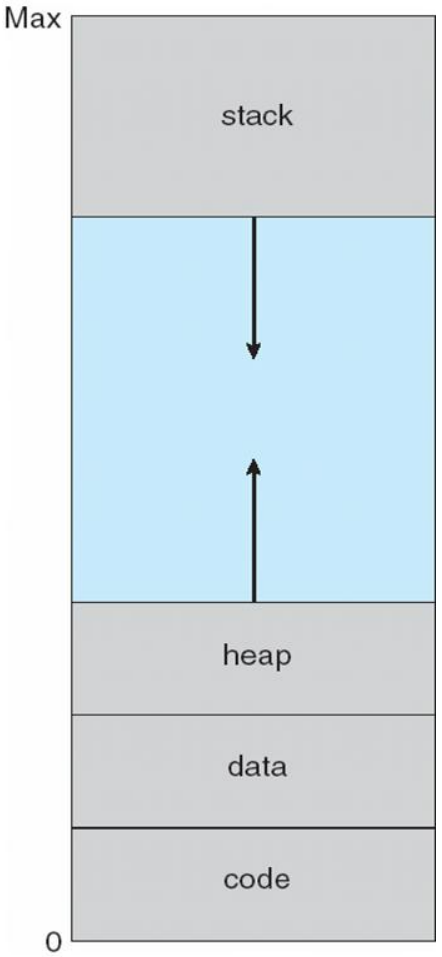
- **Virtual address space** refers to logical view of how process is stored in memory
  - Usually start at address 0, contiguous addresses until end of space
  - Meanwhile, physical memory organized in page frames
  - MMU must map logical to physical
- Virtual memory can be implemented via:
  - Demand paging
  - Demand segmentation

Virtual Memory That is Larger Than Physical Memory

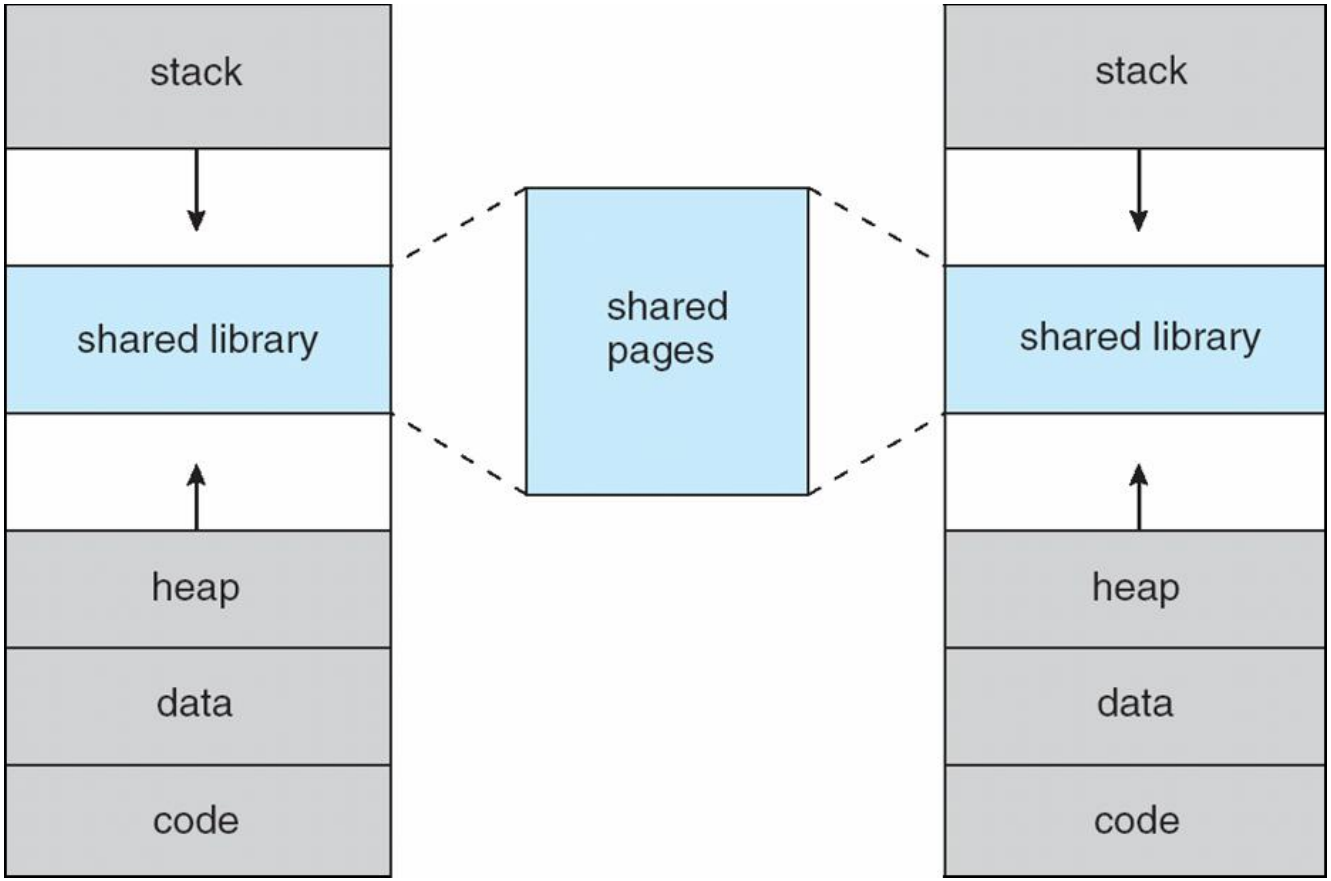


Virtual-address Space

- Usually design logical address space for stack to start at Max logical address and grow “down” while heap grows “up”. The stack is a region of memory that stores variables created by each function (including the main function) during its execution. The heap is a region of memory used for dynamic memory allocation, where blocks of memory are allocated and released in an arbitrary order
  - Maximizes address space use
  - Unused address space between the two is holes
    - No physical memory needed until heap or stack grows to a given new page
- Virtual address spaces that include holes are known as sparse address spaces.
- Enables **sparse** address spaces with holes left for growth, dynamically linked libraries, etc
- System libraries can be shared by several processes via mapping into virtual address space
- Shared memory by mapping pages read-write into virtual address space
- Pages can be shared during `fork()` system call, thus speeding up process creation.

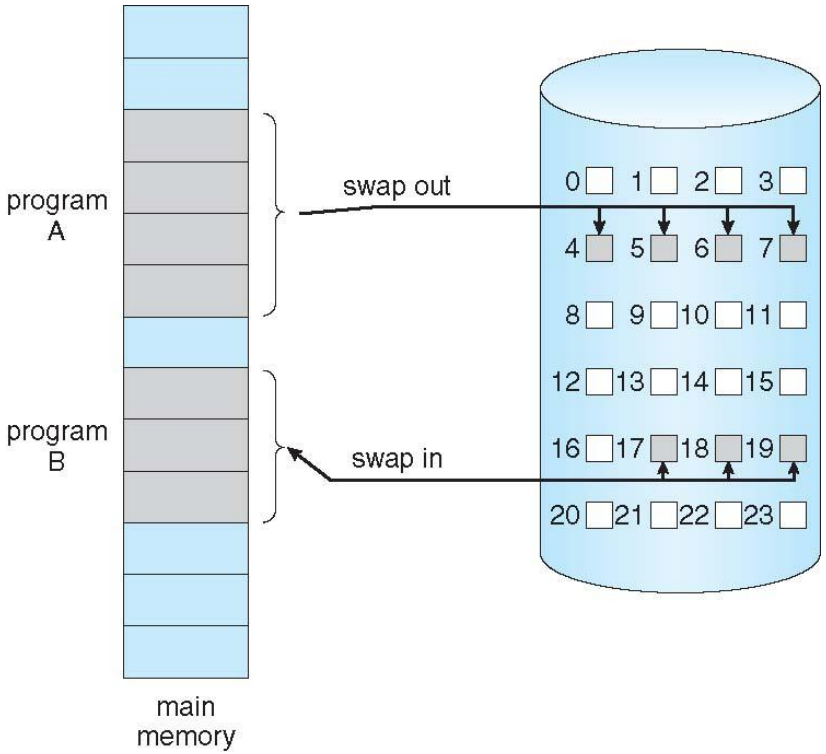


Shared Library Using Virtual Memory



Demand Paging

- In Demand paging, pages are loaded only when they are demanded during program execution.
- Could bring entire process into memory at load time
- Or bring a page into memory only when it is needed
  - Less I/O needed, no unnecessary I/O
  - Less memory needed
  - Faster response
  - More users
- Similar to paging system with swapping (diagram on right)
- Page is needed ⇒ reference to it
  - invalid reference ⇒ abort
  - not-in-memory ⇒ bring to memory
- **Lazy swapper** – never swaps a page into memory unless page will be needed
  - Swapper manipulates entire process whereas a pager is concerned with the individual pages of a process.
  - Pager – in demand paging

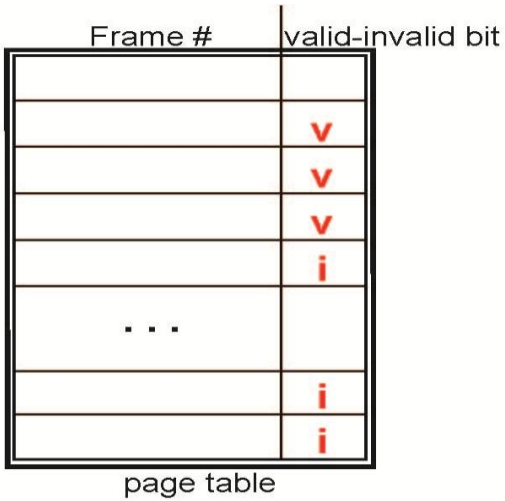


Basic Concepts

- With swapping, pager guesses which pages will be used before swapping out again
- Instead, pager brings in only those pages into memory
- How to determine that set of pages?
  - Need new MMU functionality to implement demand paging
- If pages needed are already **memory resident**
  - No difference from non demand-paging
- If page needed and not memory resident
  - Need to detect and load the page into memory from storage
    - Without changing program behavior
    - Without programmer needing to change code

Valid-Invalid Bit

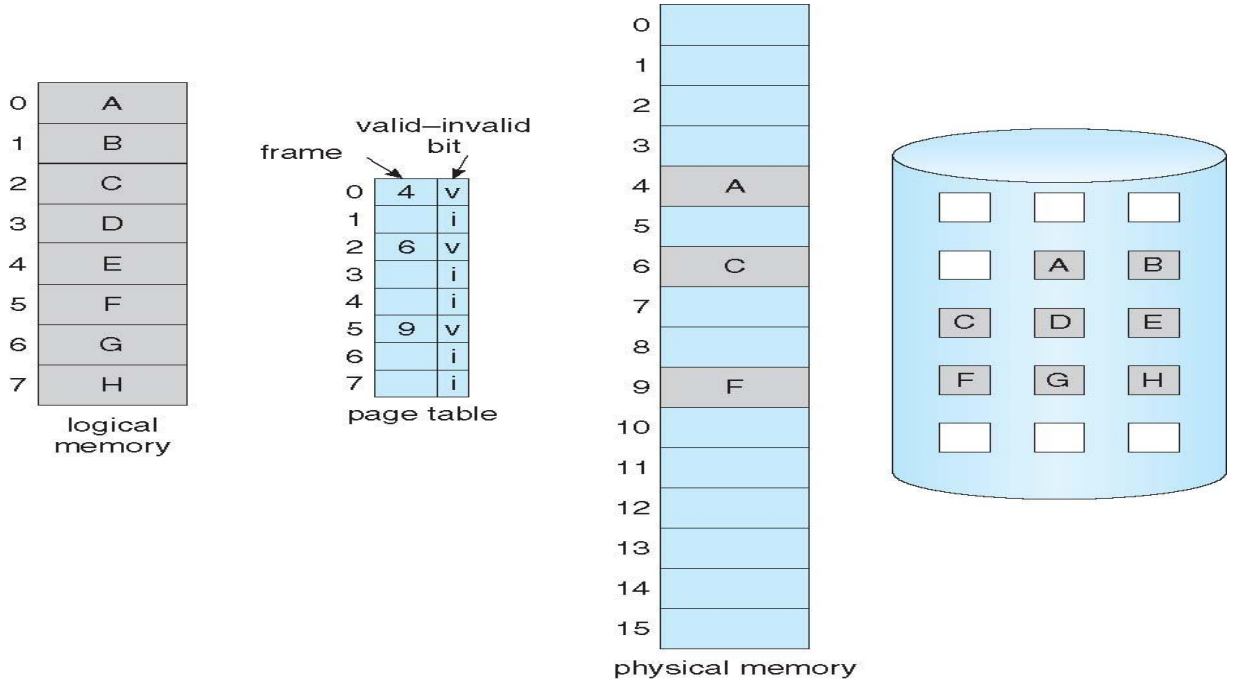
- With each page table entry a valid–invalid bit is associated (**v** ⇒ in-memory – **memory resident**, **i** ⇒ not-in-memory)
- If the bit is set to “invalid,” the page either is not valid (that is, not in the logical address space of the process) or is valid but is currently on the disk.
- Initially valid–invalid bit is set to **i** on all entries
- Example of a page table snapshot:



During MMU address translation, if valid–invalid bit in page table entry is i ⇒ page fault

Page Table When Some Pages Are Not in Main Memory

The page-table entry for a page that is brought into memory is set as usual, but the page-table entry for a page that is not currently in memory is either simply marked invalid or contains the address of the page on disk.



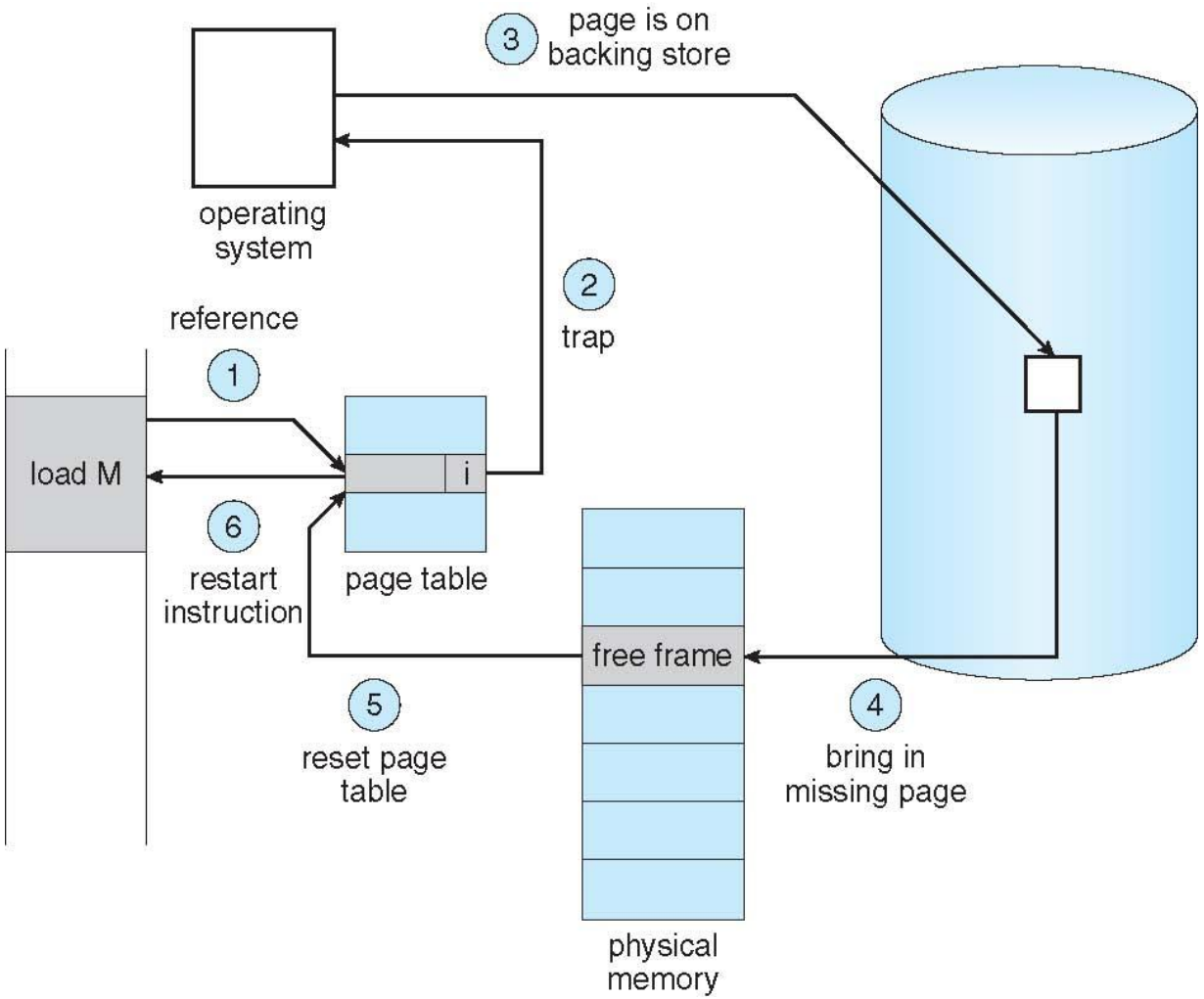
Page fault and Steps in handling page fault

1. What happens if the process tries to access a page that was not brought into memory? Access to a page marked invalid causes a **page fault**.

The procedure for handling this page fault is as follows:

1. We check an internal table (usually kept with the process control block) for this process to determine whether the reference was a valid or an invalid memory access.
2. If the reference was invalid, we terminate the process. If it was valid but we have not yet brought in that page, we now page it in.
3. We find a free frame (by taking one from the free-frame list, for example).
4. We schedule a disk operation to read the desired page into the newly allocated frame.
5. When the disk read is complete, we modify the internal table kept with the process and the page table to indicate that the page is now in memory.
6. We restart the instruction that was interrupted by the trap. The process can now access the page as though it had always been in memory.

Steps in Handling a Page Fault



Aspects of Demand Paging

- Extreme case – start process with *no* pages in memory
  - OS sets instruction pointer to first instruction of process, non-memory-resident -> page fault
  - **Pure demand paging-never bring a page into memory until it is required.**
- Actually, a given instruction could access multiple pages -> multiple page faults-unlikely to happen.
  - Consider fetch and decode of instruction which adds 2 numbers from memory and stores result back to memory
- Hardware support needed for demand paging
  - Page table with valid / invalid bit
  - Secondary memory (swap device with **swap space**)
  - Instruction restart

Performance of Demand Paging

A page fault causes the following sequence to occur:

1. Trap to the operating system
2. Save the user registers and process state
3. Determine that the interrupt was a page fault
4. Check that the page reference was legal and determine the location of the page on the disk
5. Issue a read from the disk to a free frame:
    1. Wait in a queue for this device until the read request is serviced
    2. Wait for the device seek and/or latency time
    3. Begin the transfer of the page to a free frame
6. While waiting, allocate the CPU to some other user
7. Receive an interrupt from the disk I/O subsystem (I/O completed)
8. Save the registers and process state for the other user
9. Determine that the interrupt was from the disk
10. Correct the page table and other tables to show page is now in memory
11. Wait for the CPU to be allocated to this process again
12. Restore the user registers, process state, and new page table, and then resume the interrupted instruction

Performance of Demand Paging (Cont.)

- Three major activities
  - Service the interrupt – careful coding means just several hundred instructions needed
  - Read the page – lots of time
  - Restart the process – again just a small amount of time
- Page Fault Rate $0 \le p \le 1$
  - if $p = 0$ no page faults
  - if $p = 1$, every reference is a fault
- Effective Access Time (EAT)

$$\begin{aligned} EAT = (1 - p) \times & \text{ memory access} \\ & + p \text{ (page fault overhead} \\ & \quad + \text{swap page out} \\ & \quad + \text{swap page in)} \end{aligned}$$

Demand Paging Example

- Memory access time = 200 nanoseconds
- Average page-fault service time = 8 milliseconds
- EAT = (1 – p) x 200 + p (8 milliseconds)

  = (1 – p x 200 + p x 8,000,000

  = 200 + p x 7,999,800
- If one access out of 1,000 causes a page fault, then

  EAT = 8.2 microseconds.

  This is a slowdown by a factor of 40!!
- If want performance degradation < 10 percent
  - 220 > 200 + 7,999,800 x p
    20 > 7,999,800 x p
  - p < .0000025
  - < one page fault in every 400,000 memory accesses

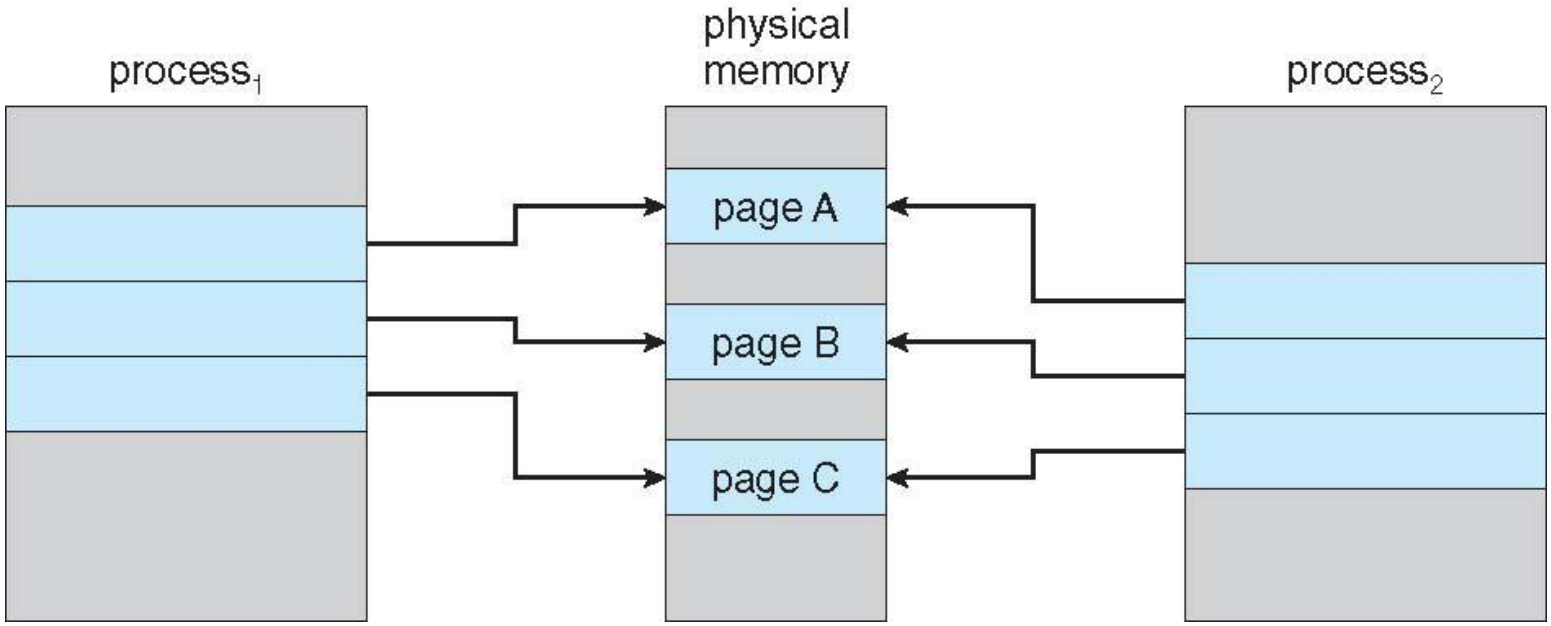
Demand Paging Optimizations

- Swap space I/O faster than file system I/O even if on the same device
  - Swap allocated in larger chunks, less management needed than file system
- Copy entire process image to swap space at process load time
  - Then page in and out of swap space
  - Used in older BSD Unix
- Demand page in from program binary on disk, but discard rather than paging out when freeing frame
  - Used in Solaris and current BSD
  - Still need to write to swap space
    - Pages not associated with a file (like stack and heap) – **anonymous memory**
    - Pages modified in memory but not yet written back to the file system
- Mobile systems
  - Typically don't support swapping
  - Instead, demand page from file system and reclaim read-only pages (such as code)

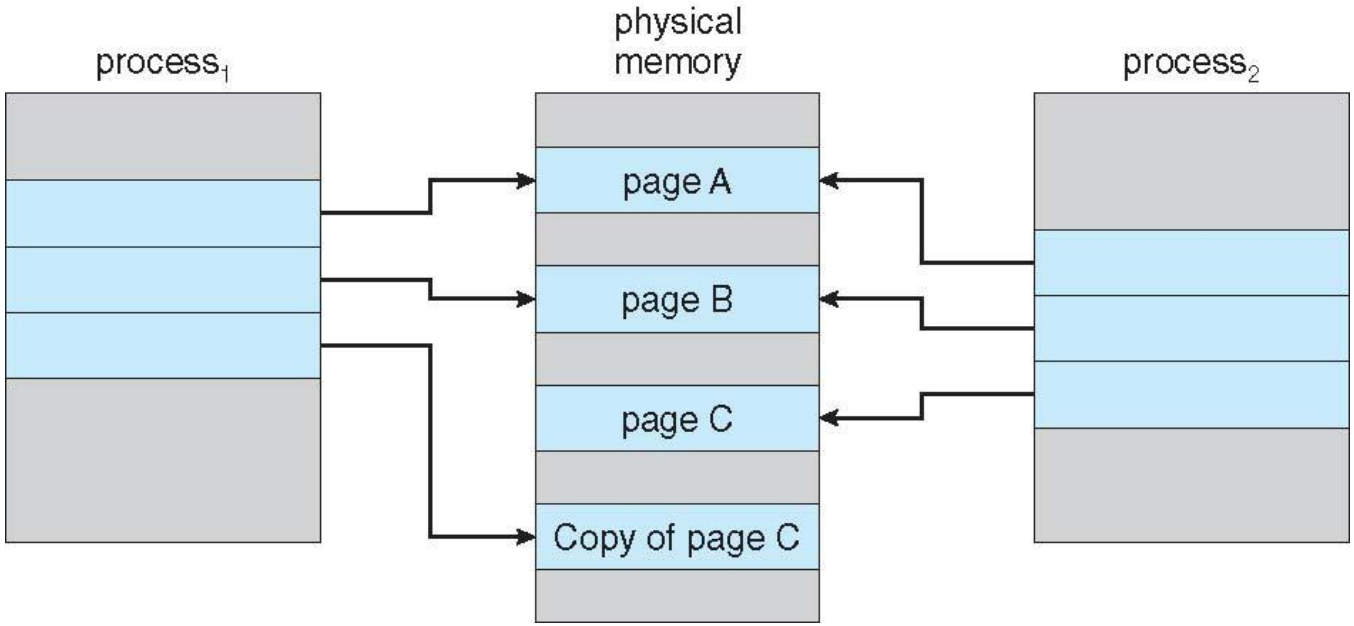
Copy-on-Write

- **Copy-on-Write** (COW) *technique allows both parent and child processes to initially **share** the same pages in memory*
  - If either process modifies a shared page, only then is the page copied
  - shared pages are marked as copy-on-write pages
  - if either process writes to a shared page, a copy of the shared page is created.
- COW allows more efficient process creation as only modified pages are copied
- In general, free pages are allocated from a **pool** of **zero-fill-on-demand** pages
  - Pool should always have free frames for fast demand page execution
    - Don’t want to have to free a frame as well as other processing on page fault

Before Process 1 Modifies Page C



After Process 1 Modifies Page C



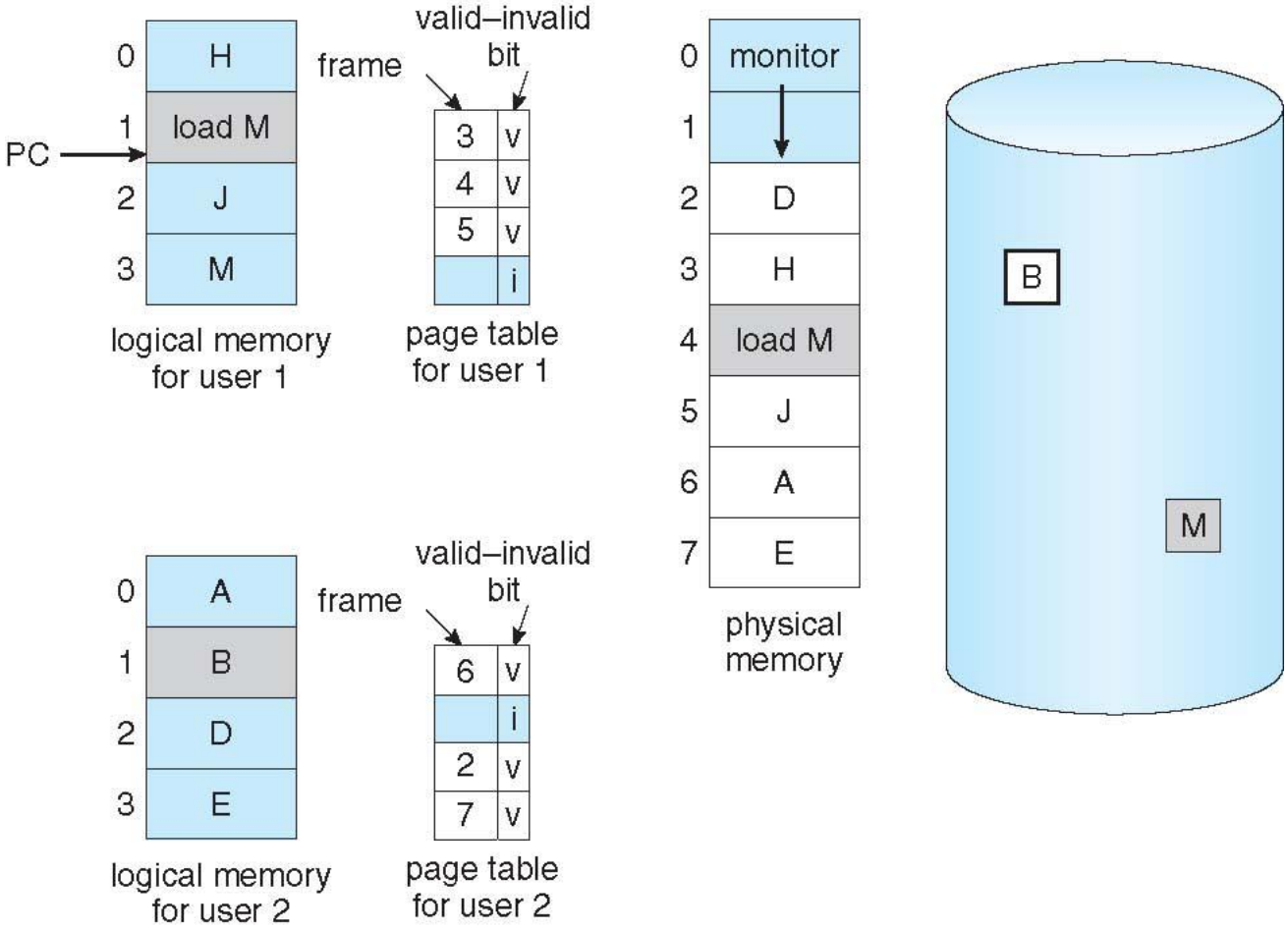
What Happens if There is no Free Frame?

- Used up by process pages
- Also in demand from the kernel, I/O buffers, etc
- How much to allocate to each?
- Page replacement – find some page in memory, but not really in use, page it out
  - Algorithm – terminate? swap out? replace the page?
  - Performance – want an algorithm which will result in minimum number of page faults
- Same page may be brought into memory several times

Page Replacement

- Prevent **over-allocation** of memory by modifying page-fault service routine to include page replacement
- Use **modify** (**dirty**) **bit** to reduce overhead of page transfers – only modified pages are written to disk
- Page replacement completes separation between logical memory and physical memory – large virtual memory can be provided on a smaller physical memory

Need For Page Replacement



Basic Page Replacement

1. Find the location of the desired page on disk

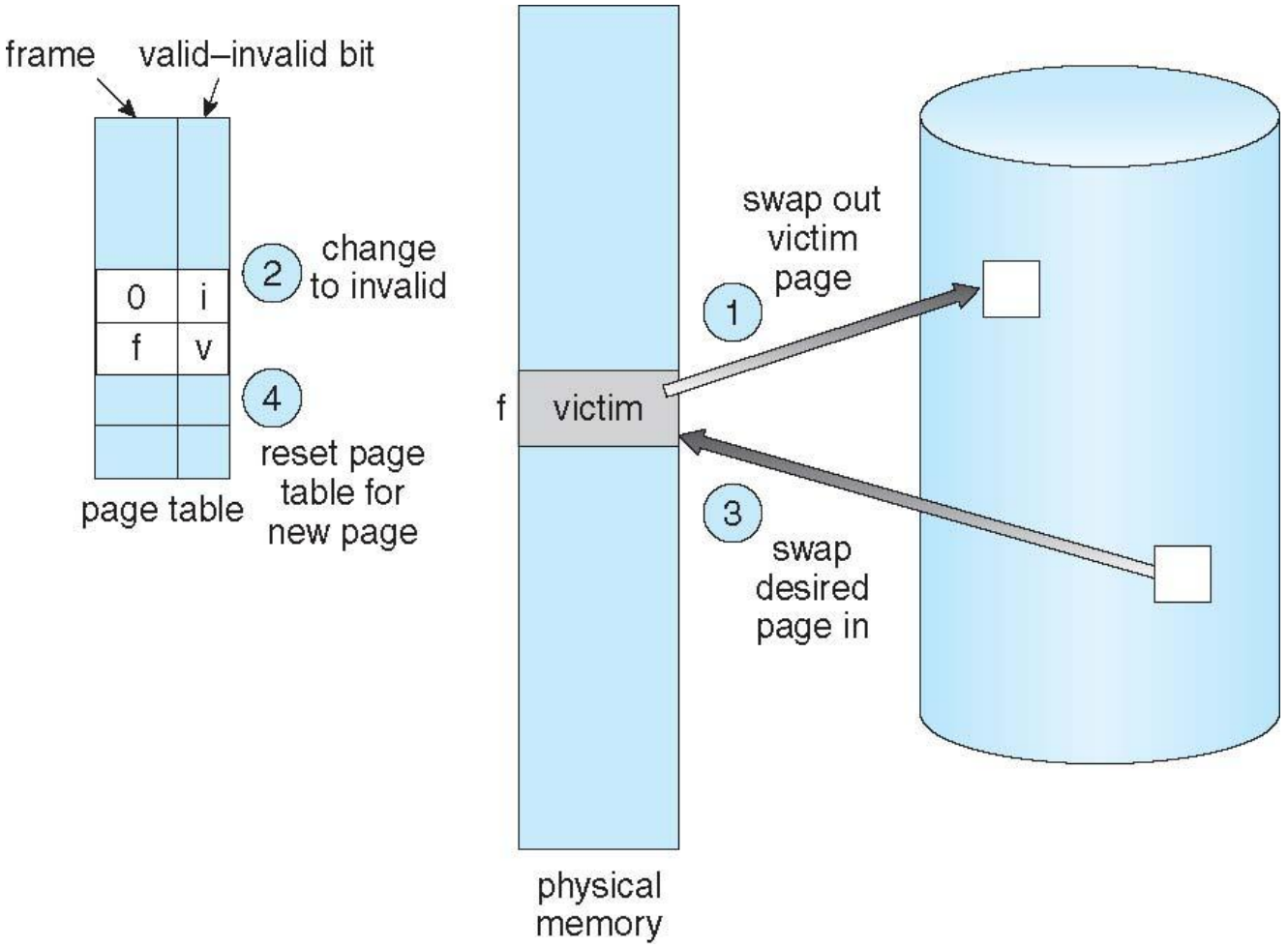
2. Find a free frame:
   - If there is a free frame, use it
   - If there is no free frame, use a page replacement algorithm to select a **victim frame**
     - Write victim frame to disk if it is dirty(modified)

3. Bring the desired page into the (newly) free frame; update the page and frame tables

4. Continue the process by restarting the instruction that caused the trap

Note now potentially 2 page transfers for page fault – increasing EAT

Page Replacement

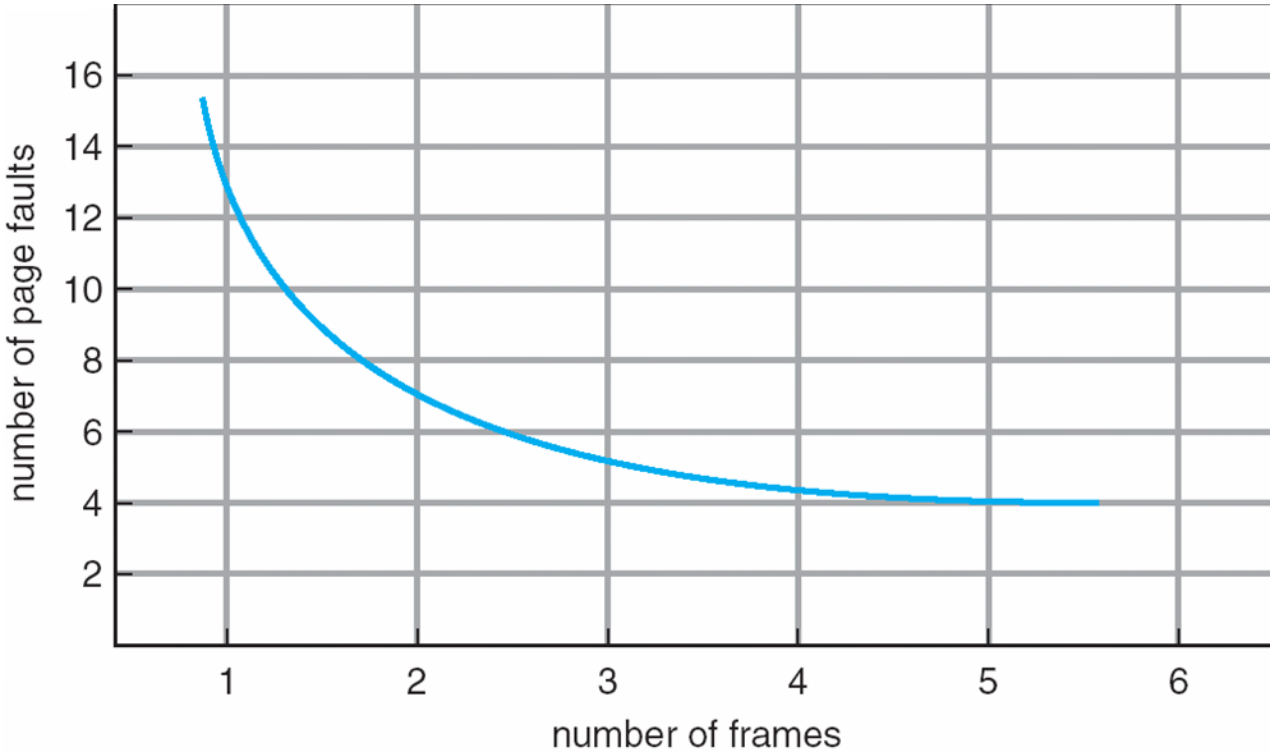


Page and Frame Replacement Algorithms

- **Frame-allocation algorithm** determines
  - How many frames to give each process
  - Which frames to replace
- **Page-replacement algorithm**
  - Want lowest page-fault rate on both first access and re-access
- Evaluate algorithm by running it on a particular string of memory references (reference string) and computing the number of page faults on that string
  - String is just page numbers, not full addresses
  - Repeated access to the same page does not cause a page fault
  - Results depend on number of frames available
- In all our examples, the **reference string** of referenced page numbers is

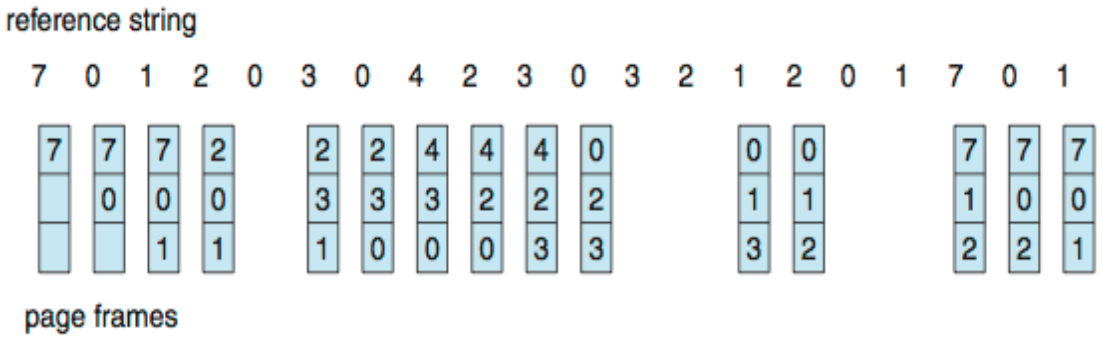
7,0,1,2,0,3,0,4,2,3,0,3,0,3,2,1,2,0,1,7,0,1

Graph of Page Faults Versus The Number of Frames



First-In-First-Out (FIFO) Algorithm

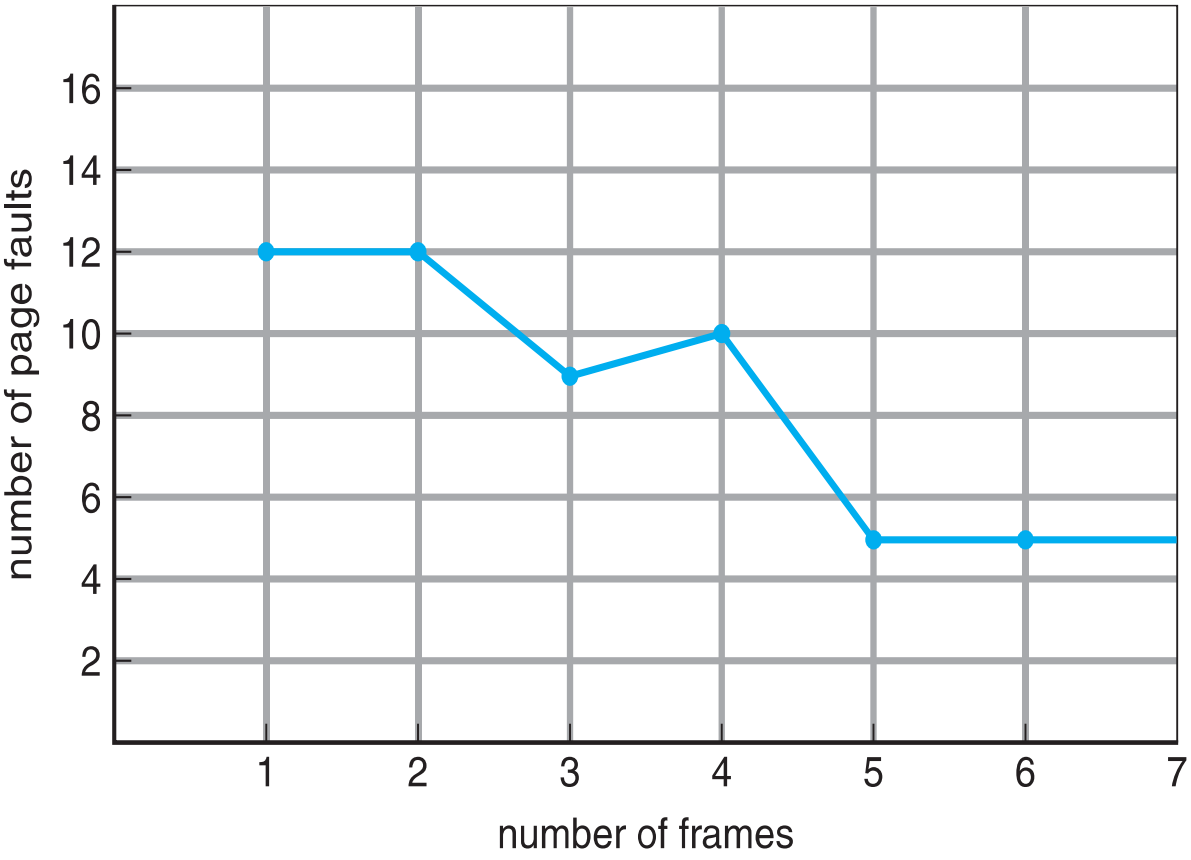
- Reference string: **7,0,1,2,0,3,0,4,2,3,0,3,0,3,2,1,2,0,1,7,0,1**
- 3 frames (3 pages can be in memory at a time per process)



15 page faults

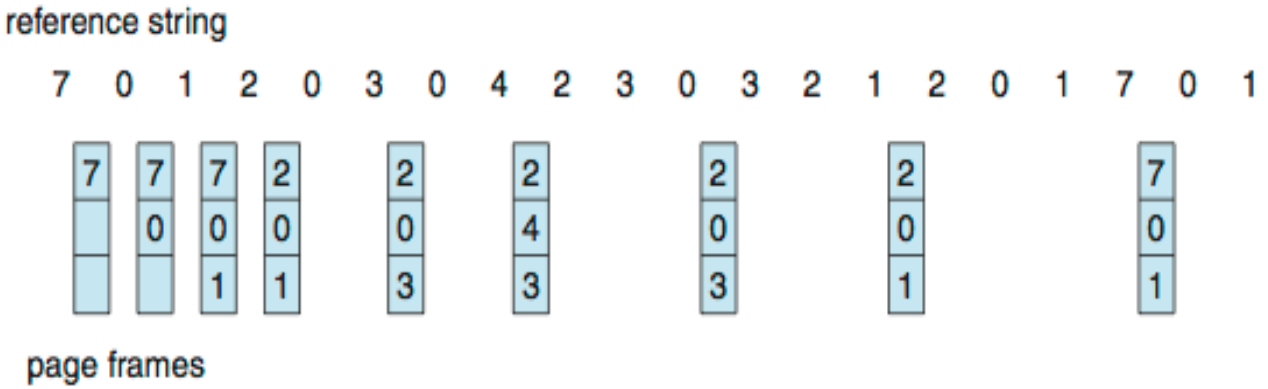
- Can vary by reference string: consider 1,2,3,4,1,2,5,1,2,3,4,5
  - Adding more frames can cause more page faults!
    - **Belady's Anomaly**
- How to track ages of pages?
  - Just use a FIFO queue

FIFO Illustrating Belady's Anomaly



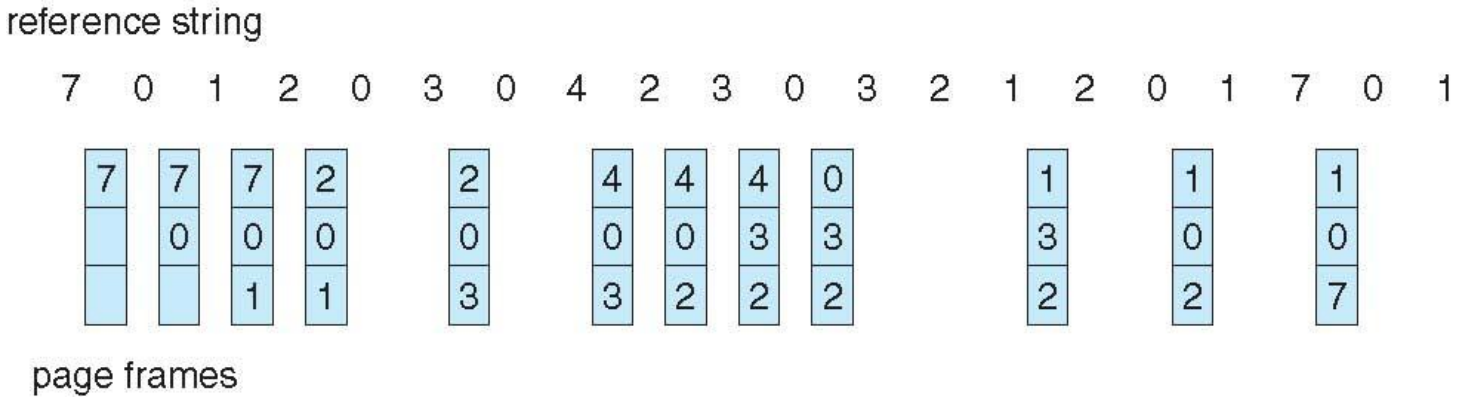
Optimal Algorithm

- Replace page that will not be used for longest period of time
  - 9 is optimal for the example
- How do you know this?
  - Can't read the future
- Used for measuring how well your algorithm performs



Least Recently Used (LRU) Algorithm

- Use past knowledge rather than future
- Replace page that has not been used in the most amount of time
- Associate time of last use with each page

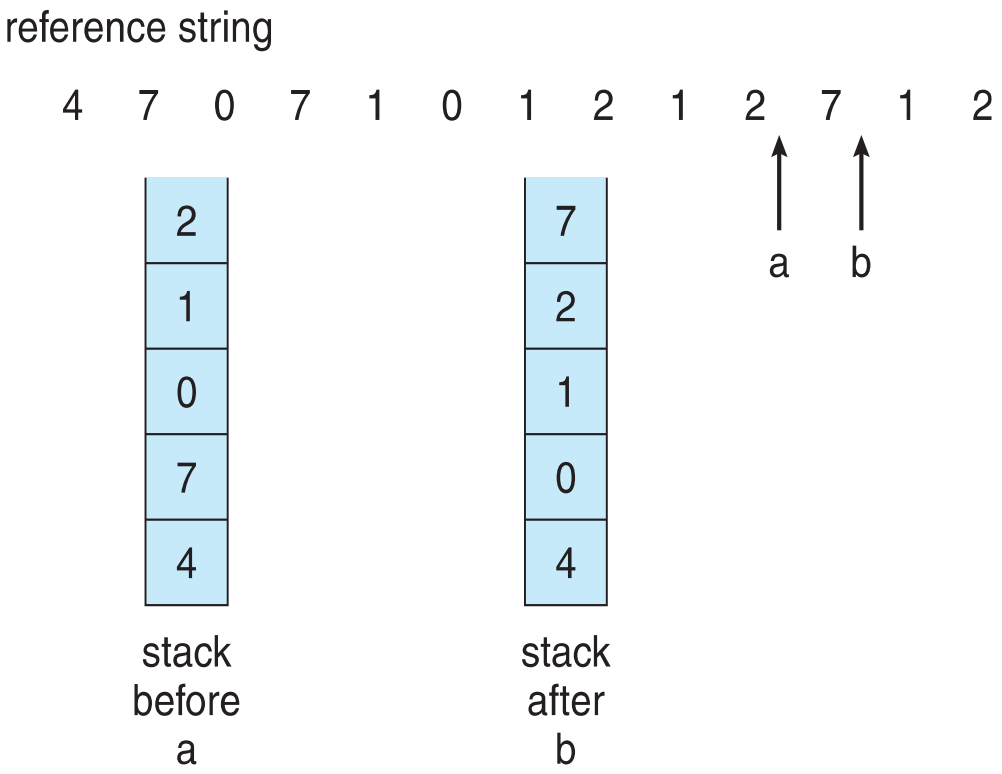


- 12 faults – better than FIFO but worse than OPT
- Generally good algorithm and frequently used
- But how to implement?

LRU Algorithm (Cont.)

- **Counter implementation**
  - Every page entry has a counter; every time page is referenced through this entry, copy the clock into the counter
  - When a page needs to be changed, look at the counters to find smallest value
    - Search through table needed
- **Stack implementation**
  - Keep a stack of page numbers in a double link form:
  - Page referenced:
    - move it to the top(thus most recently used page is always at the top and least recently used page is at bottom)
    - requires 6 pointers to be changed
  - But each update more expensive
  - No search for replacement
- LRU and OPT are cases of **stack algorithms** that don't have Belady's Anomaly

Use Of A Stack to Record Most Recent Page References



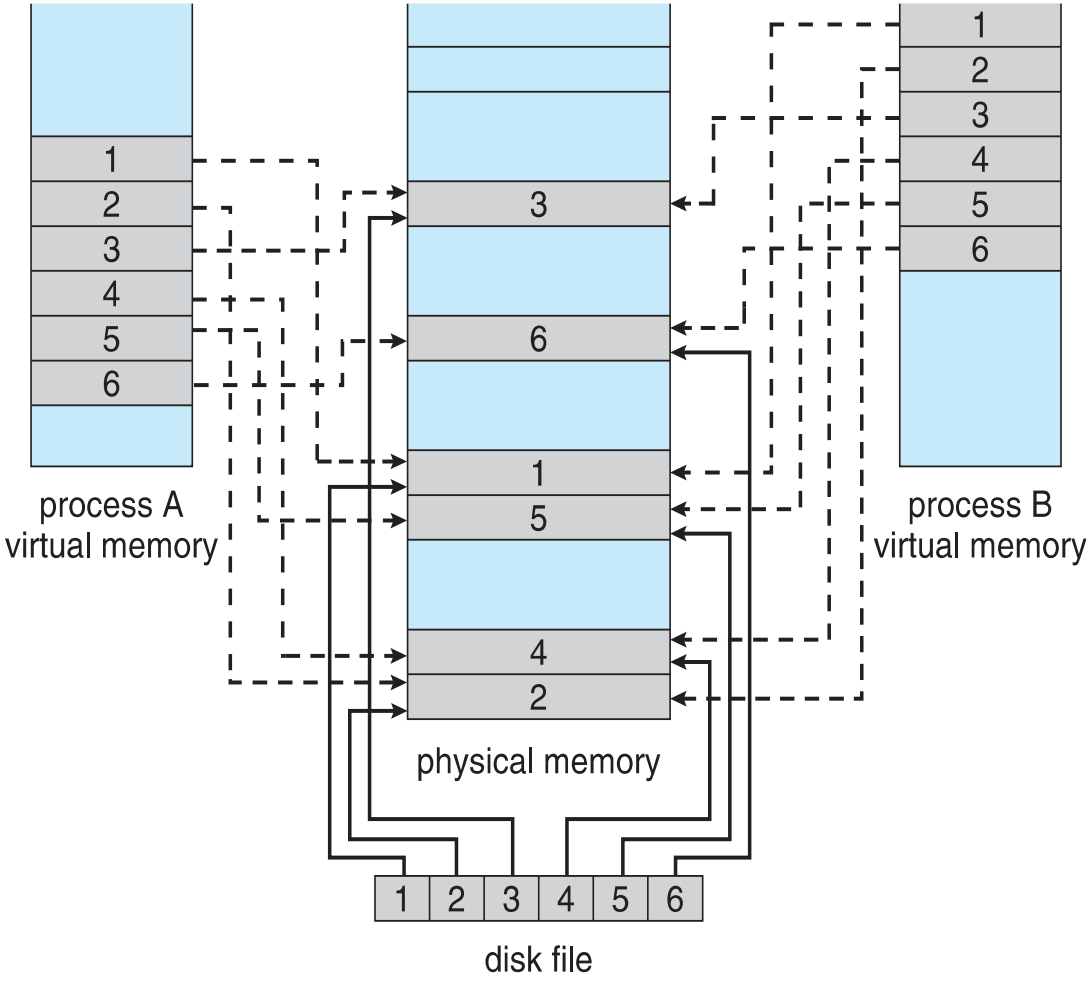
Memory-Mapped Files

- Memory-mapped file I/O allows file I/O to be treated as routine memory access by **mapping** a disk block to a page in memory
- A file is initially read using demand paging
  - A page-sized portion of the file is read from the file system into a physical page
  - Subsequent reads/writes to/from the file are treated as ordinary memory accesses
- Simplifies and speeds file access by driving file I/O through memory rather than `read()` and `write()` system calls
- Also allows several processes to map the same file allowing the pages in memory to be shared
- But when does written data make it to disk?
  - Periodically (autosave) and / or at file `close()` time
  - For example, when the pager scans for dirty pages

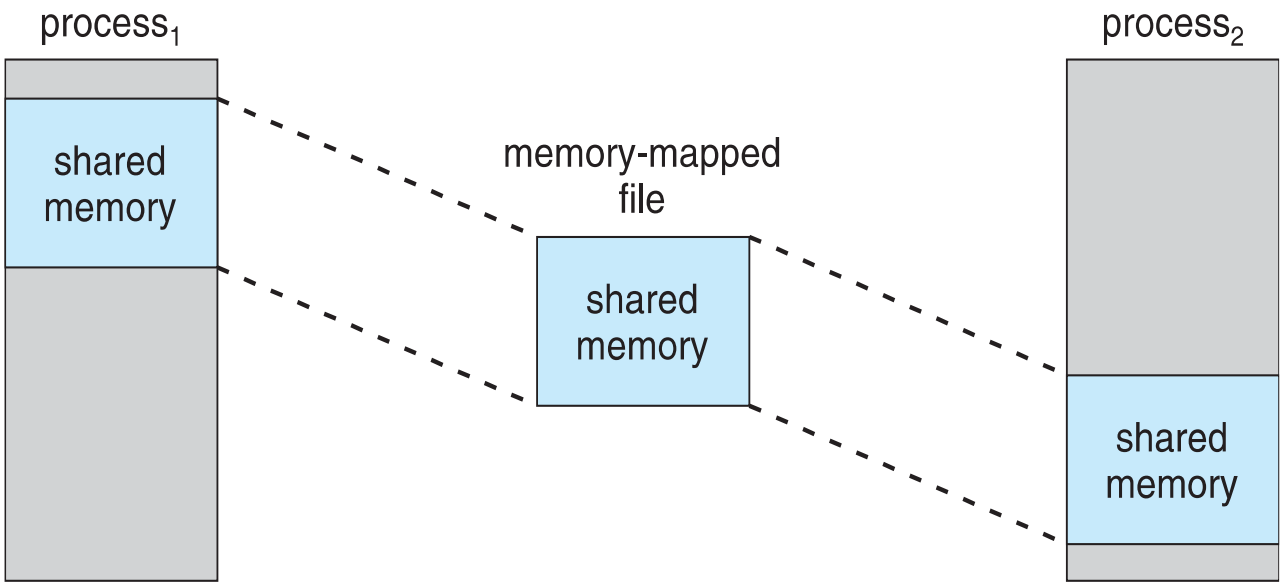
Memory-Mapped File Technique for all I/O

- Some OSes uses memory mapped files for standard I/O
- Process can explicitly request memory mapping a file via `mmap()` system call
  - Now file mapped into process address space
- For standard I/O (`open(), read(), write(), close()`), mmap anyway
  - But map file into kernel address space
  - Process still does read() and write()
    - Copies data to and from kernel space and user space
  - Uses efficient memory management subsystem
    - Avoids needing separate subsystem
- COW can be used for read/write non-shared pages
- Memory mapped files can be used for shared memory (although again via separate system calls)

Memory Mapped Files



Shared Memory via Memory-Mapped I/O



Quiz

Question 1

Consider the following page reference string:

7, 2, 3, 1, 2, 5, 3, 4, 6, 7, 7, 1, 0, 5, 4, 6, 2, 3, 0 , 1.

Assuming demand paging with three frames, how many page faults would occur for the following replacement algorithms?

1. LRU replacement
2. FIFO replacement
3. Optimal replacement

Quiz

Question 2

Consider six memory partitions of size 200 KB, 400 KB, 600 KB, 500 KB, 300 KB and 250 KB. These partitions need to be allocated to four processes of sizes 357 KB, 210 KB, 468 KB and 491 KB in that order.

Perform the allocation of processes using:

1. First Fit Algorithm
2. Best Fit Algorithm
3. Worst Fit Algorithm

Question 3 :

Given free memory partitions of 100K, 500K, 200K, 300K, and 600K (in order), how would each of the First-fit, Best-fit, and Worst-fit algorithms place processes of 212K, 417K, 112K, and 426K (in order)? Which algorithm makes the most efficient use of memory?